



Cairo University
Faculty of Economics and Political Science
Statistics Department
Arabic Section



THE DETERMINANTS OF FINANCIAL INCLUSION IN EGYPT: (2021 and 2024)

Graduation Project – Spring 2026

Authors

Nada Mohamed Abbas	Shrouk Ekramy Anter
Nada Hussein Abdullah	Yomna Mukhtar Ali
Nancy Yasser Elshamy	

Supervisor: Dr. Wafaa Ibrahim

Abstract

This study examines financial inclusion in Egypt between 2021 and 2024, focusing on the role of digital transformation, socioeconomic inequality, and behavioral mechanisms in shaping financial access and financial resilience. Using data from the World Bank Global Findex, the study applies logistic regression, clustering analysis, decomposition methods, and structural equation modelling to provide a comprehensive assessment of financial inclusion dynamics.

The results show a clear improvement in financial inclusion over time, driven mainly by the expansion of digital financial services and government-led digitization policies. However, this progress remains uneven, with persistent disparities across income groups, education levels, and geographic areas. Economic vulnerability continues to be the main barrier to inclusion, while internet access and education emerge as the strongest determinants of financial participation.

The findings also reveal a structural shift toward a more digitally connected but financially excluded group, alongside a widening urban–rural digital divide. Furthermore, the structural equation model indicates that financial inclusion improves financial resilience indirectly through saving behavior rather than through a direct effect.

Overall, the study concludes that financial inclusion in Egypt is shifting from simple access to financial services toward effective usage and behavioral integration. Policy efforts should therefore combine digital infrastructure expansion with interventions that enhance financial capability and promote saving behaviors to achieve inclusive financial development.

Keywords: Financial inclusion, digital transformation, Egypt, financial resilience, inequality, structural equation modelling

Acknowledgement

We would like to express our sincere gratitude to Prof. Fatma El-Zanaty and Dr. Mennat Allah G. Abd Elnasser for their valuable support, encouragement, and assistance throughout this work. Their kindness, guidance, and cooperation are truly appreciated.

To our esteemed supervisor, Prof. Wafaa Ibrahim, for her invaluable guidance, continuous support, and dedication throughout this research journey. Her insightful feedback and encouragement have been instrumental in shaping and strengthening this work. We are also deeply grateful for having had the opportunity to learn from her across four courses, during which she has always been a true source of knowledge and inspiration. We conclude this learning journey with sincere appreciation and a touch of sadness.

To our respected faculty and the distinguished institution to which we are proud to belong, for providing us with the academic environment and opportunities that made this research possible.

To our beloved families, who have always been the foundation of our strength and motivation.

A lovely dedication from Shrouk Ekramy to her grandfather, Antar Abdel Hady, in appreciation of his love, generosity, and support.

A special dedication from Yomna to her beloved mother, Ragia Taha, in appreciation of her unconditional support and continuous encouragement.

And a heartfelt dedication from Nada Abbas to her father, Mohamed Abbas, her mother, Wafaa Morsy, and her brother, Abbas Mohamed, in recognition of their constant love, support, and belief in her throughout this journey.

A heartfelt dedication from Nada Hussein to her beloved family: her mother, Mona Hammad, her father, Hussein Abdullah, her sisters, Radwa Hussein and Marwa Hussein, and her brothers, Ahmed Hussein and Abdullah Hussein, for their endless love, support, and encouragement throughout every step of this journey.

A loving dedication from Nancy Elshamy to her mother, Rania Samir, her father, Yasser Elshamy, her brother, Nour Elshamy, her aunt, Reham Samir, her grandmother, Rabha Hamed, and her school friends, whose love, care, prayers, and constant presence have been the greatest source of strength and support throughout this journey.

Table Of Contents

CHAPTER (1) Introduction.....	7
Introduction.....	8
Literature Review.....	9
Conceptual framework.....	12
Study Problem.....	12
Data Collection	12
Objectives	13
Methodology.....	14
I. Contingency Tables and Data Visualization.....	14
II. Fairlie Decomposition	14
III. Clustering Analysis.....	15
IV. Binary Logistic Regression	16
V. Structural Equation Modeling (SEM).....	17
CHAPTER (2) Data Preparation & Descriptives.....	18
Data Preparation.....	19
I. Data Processing and Missing Data Treatment.....	19
II. Final Selection of Analytical Variables	19
III. Data Recoding & Harmonization	19
IV. Construction of Financial Resilience.....	21
V. Construction of Pre- and Post-COVID Financial Indicators	21
VI. Construction of Financial Exclusion Barrier Variables.....	22
Univariate Analysis.....	22
Bivariate Analysis.....	25
Chapter (3) Data Analysis.....	27
Digital Access and Urban–Rural Disparities	29
I. Variable Construction and Recoding.....	29
II. Model Specification.....	29
III. Multicollinearity Diagnostics	29
IV. Logistic Regression	30
V. Results for 2021’s Model:	30
VI. Results of 2024 Model.....	31
VII. Models Performance.....	31
VIII. Fairlie Decomposition	32
IX. Variable Contributions for Digital Divide.....	32
Financial Usage Profiles in the Egyptian Population.....	34
I. Model Construction	34
II. Analysis for 2021.....	35
III. Analysis for 2024.....	36
IV. Evolution of Financial and Digital Inclusion Clusters in Egypt (2021–2024)	37

Financial Inclusion, Relative Poverty, and Their Determinants	37
I. The Initial Access Model.....	38
II. The Conditional Effects Model	40
III. Relative Poverty Determinants (Reverse Model).....	42
IV. Summary.....	44
From Access to Resilience: Financial Inclusion Pathways in Egypt	45
I. Outcome Variable: Financial Resilience vs. Financial Worry	45
II. Exclusion of the Borrowing Variable	46
III. Theoretical Path Structure	46
IV. Model Fit	47
V. Structural Path Coefficients.....	47
VI. Summary.....	50
Chapter (4) Conclusion	52
Discussion	53
I. Overview of Financial Inclusion Trends	53
II. Economic Vulnerability as the Primary Barrier	53
III. The Persistence of Income-Based Inequality (Most Important Finding).....	53
IV. Bidirectional Relationship Between Financial Inclusion and Socioeconomic Position	53
V. Education and Internet Access as Determinants of Financial Inclusion.....	54
VI. The Widening Urban–Rural Digital Divide	54
VII. The Digitally Connected Excluded: A New Policy Challenge.....	54
VIII. Stagnant Saving and Rising Borrowing.....	54
IX. The Limited Role of Merchant Digital Payments.....	55
X. Evolving Barriers: Documentation and Trust.....	55
XI. COVID-19 and Behavioral Change: An Exploratory Finding	55
XII. Gender, Employment, and Financial Inclusion: Women Benefit More from Employment.....	55
XIII. From Access to Resilience: The Mediating Role of Saving.....	55
Final Recommendations.....	56
References.....	58
Appendix A: Data Codebook and Variable Description.....	60
Appendix B: Model Specification and Diagnostic Tests	63
Appendix C: Structural Equation Modelling Estimation and Model Assumptions	63
Appendix D: Spline Specification in Logistic Regression Models.....	65
Appendix E: Mathematical Formulation of the Fairlie Decomposition Framework	66

List Of Figures

Figure 2.1: Distribution of Egypt's Population Across Income Quintiles (2021–2024).....	22
Figure 2.2: Comparison of Financial Resilience Between 2021 and 2024.....	23
Figure 2.3: The Digital Transformation of Government Transfers in Egypt (2021–2024)	24
Figure 2.4: Financial Exclusion Barriers among the Unbanked in Egypt (2021 vs 2024)	24
Figure 2.5: Perceived Financial Hardship Due to COVID-19 (Egypt - 2021)	24
Figure 3.1: Decomposition of Urban–Rural Internet Access Gap by Socioeconomic Variables (2021–2024)	33
Figure 3.2: Elbow Plot and Dendogram – 2021	35
Figure 3.3: Elbow Plot and Dendogram – 2024	36
Figure 3.4: Probability of Account Ownership by Education Level.....	38
Figure 3.5: The ROC Curve of The Initial Access Model	39
Figure 3.6: The ROC Curve of The Conditional Effects Model.....	40
Figure 3.7: Probability of Being in the Upper Income Quantile Group by Education Level	42
Figure 3.8: The ROC Curve of the Relative Poverty Determinants Model.....	43
Figure 3.9: Structural Equation Modeling (SEM) framework of financial inclusion and its effects on financial resilience through savings.....	51

List Of Tables

Table 1.1: Classification Performance Measures in Logistic Regression Models (Hosmer, Lemeshow & Sturdivant, 2013).....	16
Table 1.2: Goodness-of-Fit Indices and Recommended Thresholds for SEM Evaluation (Hu & Bentler, 1999)	17
Table 2.1: 2021 Recoding Summary.....	20
Table 2.2: 2024 Recoding Summary.....	20
Table 2.3: Harmonization Actions	21
Table 2.4: Demographic and Educational Characteristics in Egypt (2021 vs 2024)	22
Table 2.5: Financial Inclusion, Behavior, and Digital & Access Indicators in Egypt (Percentage of “Yes” Responses, 2021 vs 2024).....	23
Table 2.6: Payment Disruptions Across Contexts During COVID-19 (Egypt – 2021)	25
Table 2.7: Demographic Characteristics and Account Ownership (Egypt, 2021 vs 2024).....	25
Table 2.8: Financial Behavior (Saving and Borrowing) and Account Ownership (Egypt, 2021 vs 2024).....	26
Table 2.9: Financial Resilience and Account Ownership (2021 vs 2024)	26
Table 2.10: Internet Access, Urban–Rural Disparities, and Financial Inclusion in Egypt (2021 - 2024).....	26
Table 3.1: Multicollinearity Diagnostics for Urban and Rural Logistic Regression Models in Egypt (2021–2024)	29
Table 3.2: Logistic Regression Results for Determinants of Internet Access in Egypt (2021)	30
Table 3.3: Logistic Regression Results for Determinants of Internet Access in Egypt (2024)	31
Table 3.4: Fairlie Decomposition of Urban–Rural Gap in Internet Access (2021 vs 2024).....	32
Table 3.5: 2021 Cluster Profiles.....	35
Table 3.6: 2024 Cluster Profiles.....	36
Table 3.7: Odds Ratio of The Initial Access Model	39
Table 3.8: Odds Ratio of The Conditional Effects Model	41
Table 3.9: Odds Ratio of Relative Poverty Determinants Model	43
Table 3.10: Structural Equation Model Fit Statistics	47
Table 3.11: Equation (1) Determinants of Account Ownership (Financial Inclusion)	48
Table 3.12: Equation (2) Determinants of Saving Behavior.....	48
Table 3.13: Equation (3) Determinants of Financial Resilience	49
Table 3.14: Indirect Effect of Account Ownership on Financial Resilience via Saving.....	49
Table 4.1: Summary of Empirical Findings Across Methods	56

CHAPTER (1)

Introduction

Introduction

In recent years, financial inclusion has emerged as a central pillar of Egypt's economic reform agenda and its sustainable development strategy (Egypt Vision 2030). Beyond the expansion of traditional banking services, it is increasingly recognized as a key mechanism for enhancing economic welfare, reducing inequality, and strengthening macroeconomic stability through the integration of individuals into the formal financial system.

This study adopts the conceptual framework of the World Bank Global Findex database to ensure international comparability and consistency in measuring financial inclusion. Within this framework, financial inclusion refers to the access to and usage of formal financial services, including savings, payments, credit, and digital financial products provided by regulated financial institutions (Demirgüç-Kunt et al., 2022).

More specifically, account ownership is defined as having an individually or jointly held account at a regulated financial institution, such as a bank, credit union, microfinance institution, post office, or mobile money provider. It also includes the active use of such accounts for financial transactions, including receiving wages, government transfers, pensions, agricultural payments, or paying utility bills (Demirgüç-Kunt et al., 2022).

In addition, digital financial activity refers to the use of mobile phones or the internet to conduct financial transactions, including payments, transfers, and account management. This reflects the growing importance of digital infrastructure in reshaping access to and usage of financial services. Financial behavior, in turn, captures saving and borrowing practices, reflecting how households manage resources, smooth consumption, and respond to financial shocks (Allen et al., 2016).

The period 2021 - 2024 is particularly significant due to the coexistence of major global and domestic developments. At the global level, the post-COVID-19 recovery phase was marked by accelerated digital transformation and rising inflationary pressures, which reshaped financial behavior and access to financial services (IMF, 2023). At the domestic level, Egypt implemented a series of financial and monetary reforms led by the Central Bank of Egypt, aimed at strengthening financial stability, expanding digital payment infrastructure, and promoting financial inclusion as a national priority. These reforms included the expansion of mobile banking services, the development of digital payment systems, and efforts to integrate informal economic activities into the formal financial sector.

This makes the selected period particularly suitable for assessing how structural reforms, digital adoption, and macroeconomic shocks jointly shape financial inclusion outcomes.

This study makes four key contributions to literature. First, it provides micro-level comparative evidence on financial inclusion in Egypt across two critical periods (2021 and 2024), capturing post-pandemic dynamics alongside the ongoing digital transformation. Second, it integrates multiple empirical approaches - including logistic regression, Fairlie decomposition, clustering analysis, and structural equation modelling - within a unified analytical framework.

Third, it extends the literature by conceptualizing financial inclusion as a behavioral transmission mechanism rather than a final outcome, highlighting its role in shaping financial resilience through saving behavior. Finally, it provides new evidence on the evolving role of digital access as a structural determinant of inequality, particularly through widening urban–rural. Therefore, this study empirically examines the key socioeconomic, demographic, digital, and behavioral determinants of financial inclusion in Egypt over the period 2021 - 2024, while also capturing its interaction with broader inequality and poverty dynamics.

Importantly, this study also recognizes the bidirectional relationship between financial inclusion and relative poverty. On one hand, financial inclusion can reduce relative poverty by improving access to savings, credit, and digital financial services, thereby enhancing income stability and economic resilience.

On the other hand, relative poverty can act as a barrier to financial inclusion by limiting access to formal financial services due to income constraints, lack of documentation, and low financial literacy. This dual relationship highlights the need to treat financial inclusion not only as an outcome, but also as both a driver and a consequence of poverty dynamics.

Literature Review

This literature review examines key studies on financial inclusion, focusing on its determinants, its effect on economic well-being, and the main challenges in expanding access to financial services.

Stiglitz and Weiss (1981) and Akerlof (1970) investigated the fundamental causes of financial exclusion through a theoretical economic modeling approach based on the principles of Information Economics. Their research focused on the impact of asymmetric information on the functionality of credit markets, demonstrating that market imperfections arise when lenders lack sufficient data regarding the creditworthiness of borrowers. The researchers concluded that this information gap leads to adverse selection and moral hazard, which force formal financial institutions to implement credit rationing strategies. Consequently, their findings suggest that even when low-income individuals or small-scale firms are willing to pay higher interest rates, they remain excluded from formal markets because lenders perceive them as high-risk segments that could jeopardize the stability of the financial portfolio (Akerlof, 1970; Stiglitz & Weiss, 1981).

Williamson (1985) analyzed institutional barriers to financial access using the Transaction Cost Economics (TCE) framework, focusing on the role of economic frictions in financial intermediation. His work highlighted that high transaction costs—driven by information gathering, contract enforcement, and monitoring of low-income clients—make it economically unviable for traditional banks to serve marginalized and geographically dispersed populations. He concluded that when transaction costs are high relative to the size of financial transactions, a structural barrier is created that excludes the poor from formal financial systems (Williamson, 1985).

Fu and Mishra (2021) analyzed digital financial ecosystems and found that while technological innovation reduces transaction costs, it can also create new risks such as digital exclusion and exploitative lending practices. They concluded that without strong regulation, digital finance may increase rather than reduce financial vulnerability (Fu & Mishra, 2021).

Karlan, Ratan, and Zinman (2014) employed a behavioral economics framework to critically examine the traditional assumptions of borrower rationality in financial decision-making. Their research argued that focusing solely on reducing information and cost barriers overlooks deep-seated behavioral biases and socio-cultural factors that influence household choices, even when market frictions are minimized. In a complementary vein, Atkinson and Messy (2013) conducted a demand-side assessment highlighting the limitations of supply-oriented financial models. Their findings indicated that exclusion often persists due to low financial literacy, lack of institutional trust, and unsuitable product design, emphasizing the importance of consumer-side behavioral and educational factors in inclusion strategies (Atkinson & Messy, 2013; Karlan, Ratan, & Zinman, 2014).

A recent study by Sahay et al. (2025) found that both traditional and digital financial inclusion significantly enhance financial resilience by improving households' ability to cope with economic shocks, particularly through savings and transaction accounts. Similarly, the International Monetary Fund (2023) showed that greater financial inclusion strengthens financial system resilience by reducing risks and enhancing shock absorption, although excessive credit expansion may create vulnerabilities. In addition, Liu et al. (2024) found that digital financial inclusion is associated with lower household financial vulnerability and improved capacity to manage financial shocks, thereby enhancing financial resilience.

Wagdi and Fathi (2025) utilized hierarchical regression analysis on balanced panel data from 30 Egyptian banks (2011–2022) to investigate the impact of digital financial services on banking stability. Their findings showed that digital innovation enhances financial inclusion, which in turn improves stability by increasing capital adequacy and reducing non-performing loans. Similarly, Ganna et al. (2024) applied an empirical analysis using annual data from 17 Egyptian banks (2019–2022) to examine how digital financial inclusion influences stability during periods of economic turbulence. Using the Z-Score, they found that digital financial practices strengthen banking resilience and reduce risk exposure, highlighting the importance of expanding digital access for systemic stability (Ganna et al., 2024; Wagdi & Fathi, 2025).

Mohamed (2025) employed a comparative analytical approach using traditional econometric techniques, such as Bootstrap Regression, alongside machine learning algorithms including Random Forest (RF) and Support Vector Regression (SVR) to predict the determinants of financial inclusion in Egypt. The findings showed that AI models outperform traditional methods and emphasized the importance of human development and macroeconomic stability in expanding financial access. Complementing this perspective, Alaa Elden (2024) conducted a descriptive study based on a structured survey of 193 respondents to investigate the role of digital financial inclusion in promoting inclusive growth and environmental sustainability in Egypt.

The study identified key barriers such as digital illiteracy and infrastructure gaps, while highlighting growing interest in green finance, concluding that integrating digital financial services with sustainability goals is essential for Egypt's Vision 2030 (Alaa Elden, 2024; Mohamed, 2025).

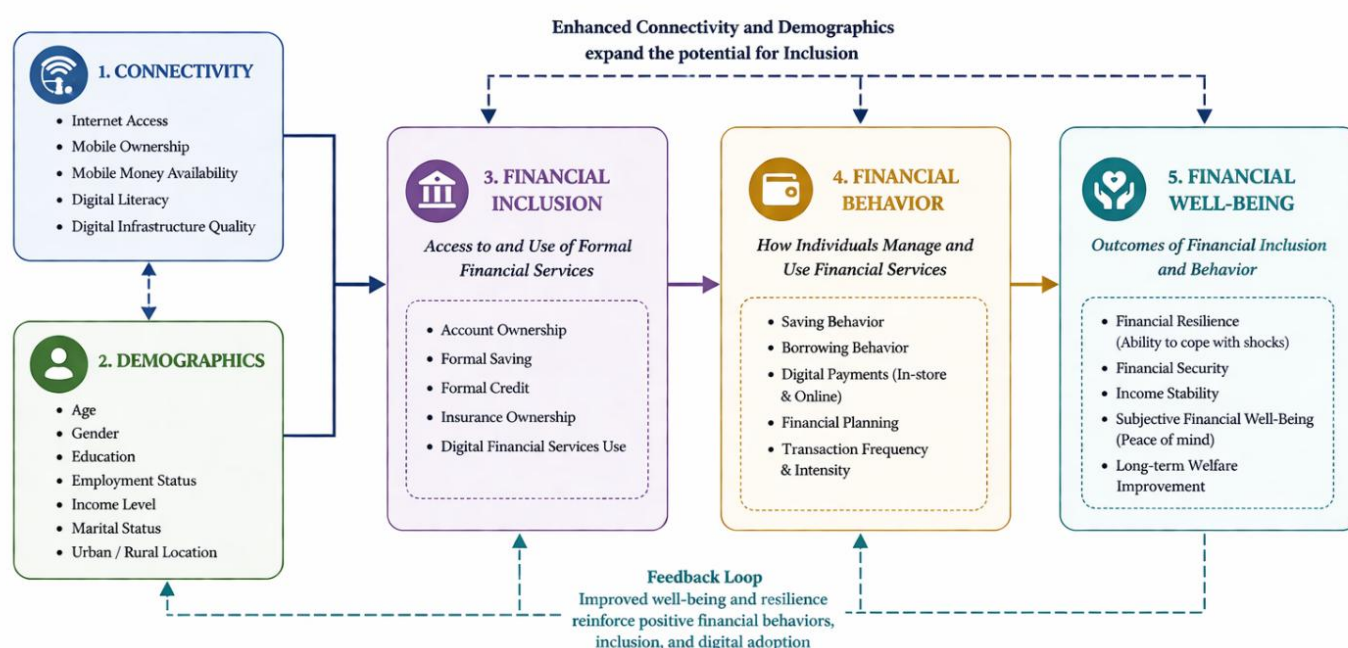
In a related contemporary context, GSMA (2026) conducted a strategic assessment of digital ecosystems to evaluate the role of Digital Financial Services (DFS) and alternative data in enhancing financial access. Their findings showed that alternative credit scoring models help address information and transaction cost barriers, enabling greater financial inclusion. The study concluded that digital innovations can reduce structural constraints in traditional financial systems and improve access for underserved groups through technology-driven solutions (GSMA, 2026).

Hegazy et al. (2026) employed a Vector Autoregression (VAR) model and descriptive analysis to examine the causal relationship between financial inclusion indicators and economic growth in Egypt from 2008 to 2023. Using semi-annual data, they analyzed delivery channels such as bank accounts, payment cards, ATMs, and credit provision. Their results showed that financial inclusion significantly affects Egypt's economic growth (GDP), emphasizing the role of digital finance as a catalyst for sustainable expansion. Similarly, Mohamed and Algarhy (2024) used logistic regression in STATA to assess the impact of financial technology on financial inclusion in Egypt (2018–2022). They found that fintech adoption significantly enhances financial inclusion and improves electronic government service efficiency, thereby improving economic welfare (Hegazy et al., 2026; Mohamed & Algarhy, 2024).

Despite extensive literature on financial inclusion, existing studies remain limited in three key aspects. First, most empirical evidence is based on macro or banking-sector data, with limited use of recent micro-level datasets capturing individual behavior in Egypt. Second, there is a lack of comparative analyses examining the evolution of financial inclusion across the post-pandemic (2021) and digital transformation (2024) periods. Third, prior research predominantly focuses on financial access rather than actual usage of digital financial services, and rarely integrates behavioral, digital, and resilience dimensions within a unified analytical framework. This study addresses these gaps by employing micro-level Findex data to examine the determinants of financial inclusion and digital usage patterns in Egypt across two time periods.

Conceptual framework

A Dynamic Pathway from Connectivity and Demographics to Financial Well-Being



This conceptual framework illustrates a progressive pathway where Connectivity and Demographics serve as the foundational drivers that enable Financial Inclusion (access to formal services), which in turn shapes Financial Behavior (how individuals manage and use those services). These behaviors directly lead to the ultimate outcome of Financial Well-Being, characterized by resilience, security, and long-term stability. The model is dynamic rather than linear, featuring a feedback loop where improved well-being reinforces positive financial habits and digital adoption, while enhanced infrastructure and demographic factors continuously expand the overall potential for inclusion.

Study Problem

Despite improvements in financial inclusion in Egypt, existing research remains limited in capturing its multidimensional and dynamic nature. Most studies focus primarily on access to financial services, with insufficient attention to usage patterns, behavioral heterogeneity, and financial resilience outcomes. Moreover, there is limited evidence on how the determinants of financial inclusion have evolved between the post-pandemic period (2021) and the phase of accelerated digital transformation (2024). Therefore, this study addresses financial inclusion as a dynamic and evolving process by examining its determinants, usage profiles, and welfare implications over time using a pooled micro-level dataset and a multi-method analytical framework.

Data Collection

This study utilizes microdata from two waves of the Global Financial Inclusion (Global Findex) Database: Global Findex 2021 and Global Findex 2024, produced by the Development Research Group, Finance and Private Sector Development Unit of the World Bank, with data collection implemented by Gallup, Inc.

The Global Findex 2021 dataset is based on nationally representative surveys covering approximately 128,000 adults across more than 120 economies. For Egypt, the sample consists of 1,003 individual-level observations collected through face-to-face interviews conducted in Arabic between September 4 and September 28, 2021. The edited public-use microdata used in this study include 114 individual-level variables. The 2021 questionnaire is structured into six key sections, capturing information on financial account ownership, saving, and borrowing behavior, digital payment usage, financial resilience, and key demographic characteristics.

The Global Findex 2024 dataset is based on nationally representative surveys of approximately 145,000 adults across 141 economies, collected between May and December 2024. For Egypt, the dataset contains 1,001 individual-level observations. The edited and anonymized dataset, accessed through a formal data request to the World Bank, includes 199 individual-level variables. Reflecting an expanded scope, the 2024 questionnaire is composed of seven sections, with the addition of a dedicated module for digital connectivity.

Both datasets employ probability-based sampling designs to ensure national representativeness. Survey weights are applied to correct for unequal probabilities of selection and nonresponse bias. The use of these two datasets enables a comparative analysis between 2021 and 2024, allowing the study to examine changes in financial inclusion indicators and assess their relationship with income position, economic resilience, and income inequality in Egypt.

Objectives

The main objective of this study is to examine the determinants of financial inclusion in Egypt and its dynamic evolution over time, through a comparative analysis of individual-level data from 2021 and 2024 using a pooled empirical framework.

Specifically, the study aims to:

- 1) Identify the key determinants of financial inclusion in Egypt and examine how socio-economic and digital factors influence individuals' access to formal financial services.
- 2) Investigate whether the effects of key determinants of financial inclusion - particularly income position across income quantiles - differ between 2021 and 2024.
- 3) Examine the direction of the relationship between income position and financial inclusion to assess whether belonging to higher income quantiles determines access to financial services, or whether financial inclusion may also contribute to upward movement in income position.
- 4) Analyze changes in payment behavior induced by the COVID-19 shock by examining shifts from cash-based transactions to digital payment methods.
- 5) Explore the relationship between financial inclusion, saving behavior, and financial resilience in order to understand how financial access translates into financial well-being.
- 6) Identify distinct financial usage profiles within the Egyptian population by examining heterogeneity in demographic characteristics and digital financial behavior.
- 7) Analyze disparities in digital access and financial inclusion across socio-economic and geographic groups and assess the factors contributing to the urban–rural digital divide.
- 8) Formulate policy recommendations based on the empirical analysis results.

Methodology

Empirical research in socio-economic and financial studies often requires a combination of complementary quantitative methods to capture different dimensions of complex phenomena. Rather than relying on a single econometric technique, a multi-method framework allows researchers to analyze determinants, behavioral mechanisms, heterogeneous patterns, and variable importance within a unified analytical structure.

I. Contingency Tables and Data Visualization

Frequency tables are used to summarize categorical variables by reporting the proportion of observations in each category (Field, 2018). Bivariate analysis is conducted to examine pairwise relationships between key variables. This is primarily implemented using cross-tabulations (contingency tables), which allow for the comparison of joint distributions across two categorical variables (Agresti, 2018). Also, conditional probabilities are computed to identify associations between binary variables such as financial inclusion status and socio-economic characteristics (Gujarati & Porter, 2009).

II. Fairlie Decomposition

(A) Analytical Framework

The analysis employs the nonlinear decomposition framework developed by Robert W. Fairlie to examine group differentials in binary outcomes. This method extends the traditional Oaxaca–Blinder decomposition to nonlinear probability models such as logit and probit regressions, making it suitable for binary dependent variables (Fairlie, 2005).

The decomposition separates the observed difference in average predicted probabilities between two groups into explained and unexplained components. The explained component reflects differences in observable characteristics, while the unexplained component captures differences in coefficients, structural relationships, behavioral responses, and unobserved heterogeneity.

Nonlinear decomposition results may also depend on the choice of reference coefficients used during decomposition. To reduce sensitivity to reference-group selection, a symmetric decomposition strategy is implemented (Shorrocks, 2013).

To enhance robustness, a resampling-with-replacement (bootstrap) procedure is implemented. The decomposition is repeatedly estimated on resampled datasets, and final results are obtained by averaging across iterations. This reduces sensitivity to sampling variability and improves stability of estimates (Fairlie, 2005).

(B) Variable-Level Decomposition

Variable-level contributions indicate each variable's role in explaining the gap: positive values widen the difference between groups, while negative values reduce it. Interpretation should consider model assumptions and potential limitations such as measurement error and omitted variable bias.

To identify the contribution of individual variables to the explained component, a sequential replacement approach is applied, where variables are introduced one at a time while holding others constant. This allows the marginal effect of each variable to be isolated within a nonlinear framework.

III. Clustering Analysis

(A) Clustering Framework

Clustering techniques are unsupervised learning methods used to partition observations into homogeneous groups based on similarity across a set of variables. These methods are widely used in exploratory data analysis to uncover hidden population structures and behavioral patterns without imposing prior assumptions on group membership.

For categorical data, K-Modes Clustering is commonly employed as an extension of K-Means, where the arithmetic mean is replaced by the mode (Huang, 1998). K-Modes is an iterative partitioning algorithm in which cluster assignments and cluster modes are repeatedly updated until convergence is achieved. It aims to minimize the within-cluster dissimilarity function defined as:

$$J = \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik} d(x_i, z_k) \quad (1.1)$$

where:

- n : number of observations
- K : number of clusters
- $\gamma_{ik} \in \{0,1\}$: cluster membership indicator
- x_i : observation vector
- z_k : cluster centroid (mode of cluster k)
- $d(\cdot)$: dissimilarity measure for categorical variables

In K-Modes, the dissimilarity between an observation and a cluster mode is defined using a simple matching dissimilarity measure, which counts the number of mismatches between categorical attributes:

$$d(x_i, z_k) = \sum_{j=1}^p \delta(x_{ij}, z_{kj}) \quad (1.2)$$

where:

- p : number of variables
- $\delta(x_{ij}, z_{kj}) = \begin{cases} 0 & \text{if } x_{ij} = z_{kj} \\ 1 & \text{otherwise} \end{cases}$

Cluster assignment is performed by allocating each observation to the cluster with the minimum dissimilarity:

$$\gamma_{ik} = \begin{cases} 1 & \text{if } d(x_i, z_k) = \min_j d(x_i, z_j) \\ 0 & \text{otherwise} \end{cases} \quad (1.3)$$

(B) Cluster Evaluation

Cluster validity is assessed using the elbow plot and dendrogram based on a dissimilarity matrix computed using Gower Distance (Gower, 1971), which is suitable for categorical and mixed-type attributes. The elbow plot is used to determine the optimal number of clusters, while the dendrogram provides a visual representation of the hierarchical structure of the data. In addition, cluster size distribution and cluster mode profiles are examined to ensure interpretability and substantive meaningfulness of the resulting segmentation.

IV. Binary Logistic Regression

(A) Analytical Framework

Binary Logistic Regression is a widely used statistical technique for modelling binary dependent variables, where the outcome takes the value 1 or 0. It estimates the probability of an event occurring as a nonlinear function of a set of explanatory variables using the logistic distribution. This approach is particularly suitable when the assumption of linearity in probability models is violated.

The model is specified as:

$$P(Y_i = 1) = \frac{1}{1 + e^{-X_i\beta}} \quad (1.4)$$

where:

- Y_i : binary outcome variable
- X_i : vector of explanatory variables
- β : vector of parameters to be estimated
- e : exponential function

Equivalently, the model can be expressed in log-odds form as:

$$\ln\left(\frac{P(Y_i = 1)}{1 - P(Y_i = 1)}\right) = X_i\beta \quad (1.5)$$

Binary logistic regression is widely used in empirical research to examine determinants of binary outcomes such as adoption, participation, access, or inclusion, particularly in social sciences and economics (Hosmer, Lemeshow & Sturdivant, 2013; Wooldridge, 2010).

(B) Model Fit Evaluation

Table 1.1: Classification Performance Measures in Logistic Regression Models (Hosmer, Lemeshow & Sturdivant, 2013)

Measure	Description	Interpretation / Role
Sensitivity (True Positive Rate)	Measures the proportion of correctly identified positive cases	Higher values (closer to 1) indicate better detection of positive outcomes
Specificity (True Negative Rate)	Measures the proportion of correctly identified negative cases	Higher values (closer to 1) indicate better identification of negative outcomes
Accuracy	Measures overall proportion of correctly classified observations	≥ 0.70 commonly considered acceptable
Youden Index	Summarizes overall diagnostic effectiveness by combining sensitivity and specificity	Used to determine the optimal cutoff point , selected at the threshold that maximizes the index

V. Structural Equation Modeling (SEM)

(A) Analytical Framework

Structural Equation Modelling (SEM) is a multivariate statistical technique used to estimate systems of simultaneous equations that capture interdependent relationships among observed variables. SEM is particularly suitable for analyzing complex behavioral processes involving multiple pathways, including direct and indirect effects. SEM is commonly estimated using maximum likelihood method. (Bollen, 1989; Kline, 2016; Hair et al., 2021)

For a single mediation pathway:

$$X \rightarrow M \rightarrow Y$$

The mediator equation calculated as:

$$M = \alpha_1 + \beta_1 X + \Gamma_1 C + \varepsilon_1 \quad (1.6)$$

The outcome equation calculated as:

$$Y = \alpha_2 + \theta_1 M + \theta_2 X + \Gamma_2 C + \varepsilon_2 \quad (1.7)$$

Where:

- X = observed predictor
- M = mediating variable
- Y = outcome variable
- C = vector of control variables

The indirect effect of the predictor is estimated using the product-of-coefficients approach (MacKinnon, 2008):

$$\text{Indirect Effect} = (\beta_1) \cdot (\theta_1) \quad (1.8)$$

This formulation allows for assessing the extent to which the effect of the predictor on the outcome is transmitted through the mediator.

(B) Model Fit Evaluation

Table 1.2: Goodness-of-Fit Indices and Recommended Thresholds for SEM Evaluation (Hu & Bentler, 1999)

Fit Index	Description	Acceptable Threshold
Chi-square test	Tests the discrepancy between the observed and model-implied covariance matrices	Non-significant preferred
Comparative Fit Index (CFI)	Compares model fit relative to an independence model	≥ 0.90 – 0.95
Tucker–Lewis Index (TLI)	Incremental fit index adjusted for model complexity	≥ 0.90 – 0.95
Root Mean Square Error of Approximation (RMSEA)	Measures approximate model fit per degree of freedom	≤ 0.06
Standardized Root Mean Square Residual (SRMR)	Measures the standardized difference between observed and predicted correlations	≤ 0.08

CHAPTER (2)

Data Preparation & Descriptives

This chapter outlines the data preparation procedures and descriptive analytical framework used in the study of financial inclusion and digital financial behavior in Egypt. It is important to ensure data consistency and cross-wave comparability, given the use of survey data from 2021 and 2024. Accordingly, a structured data processing approach was applied, including handling non-standard missing values arising from survey skip logic, harmonizing variable definitions across waves, and recoding key indicators into analytically meaningful categories. These steps ensure the reliability and comparability of the empirical analysis.

Following the data preparation stage, the chapter presents univariate and bivariate descriptive analyses to explore key patterns in financial inclusion, digital financial behavior, financial resilience, and socio-economic disparities. These descriptive statistics provide an essential foundation for understanding structural trends over time and across population groups in Egypt, and they serve as a basis for the subsequent analytical chapters of the study.

Data Preparation

I. Data Processing and Missing Data Treatment

It is important to note that the high percentage of empty observations in certain variables was not due to "Missing Data" in the traditional sense, but rather a **"Not Applicable" (N/A)** status. This occurs due to the **skip-pattern logic** of the survey, where certain questions are only asked if a specific condition is met. Additionally, an adjustment was made to the education variable by removing the incorrect fifth category, restricting it to the valid three levels (1, 2, 3) to ensure data consistency. After these adjustments, both the 2021 and 2024 datasets contain 1,001 valid observations each.

II. Final Selection of Analytical Variables

The study adopted a structured variable selection approach based on conceptual relevance, cross-wave comparability, and measurement validity. Instead of relying on individual raw questionnaire items (such as detailed fin-series questions), the analysis focuses on pre-constructed survey indicators provided within the dataset. These constructed variables were originally developed by the survey designers using multiple questionnaire responses to capture key dimensions of financial inclusion and socio-economic behavior. As a result, they offer more robust and standardized measures compared to single survey items and are not affected by item-level missing values or skip-pattern inconsistencies.

III. Data Recoding & Harmonization

a) 2021 Survey

In the 2021 dataset, variables were systematically recoded to improve interpretability and analytical consistency. Demographic variables were transformed into standardized categorical formats, while financial inclusion and digital usage variables were converted into binary indicators. Payment-related variables were simplified by collapsing multiple response categories into broader groups, and technology access variables were reduced to binary measures.

Table 2.1: 2021 Recoding Summary

Category	Variables	Recoding Outcome
Demographics	female, educ, inc_q, emp_in, urbanicity_f2f	Standard categorical groups (e.g., Female/Male, Education levels, Income quintiles, Workforce status, Rural/Urban)
Continuous	age	Kept unchanged
Financial Inclusion	account, account_fin, account_mob, saved, borrowed	Binary (Yes / No)
Digital Usage	anydigpayment, merchantpay_dig	Binary (Yes / No)
Payments	receive_wages, receive_transfers, receive_pension, receive_agriculture	Into account / Cash / Other / Did not receive
Payments	pay_utilities	Account / Cash / Other / Did not pay
Payments	remittances	Via_account, MTO_service, Cash_only, Other_method, No, Dont_know_or_refused
Technology	internetaccess, mobileowner	Binary (Yes / No)

b) 2024 Survey

The 2024 dataset followed a similar recoding framework to maintain consistency, with slight extensions to account for newly introduced digital variables. All core demographic and financial variables were processed identically to the 2021 dataset. Additional variables, such as digital account ownership, were incorporated and recoded into binary formats.

Table 2.2: 2024 Recoding Summary

Category	Variables	Recoding Outcome
Demographics	female, age, educ, inc_q, emp_in, urbanicity	Same structure as 2021
Financial Inclusion	account, account_fin, account_mob, saved, borrowed	Binary (Yes / No)
Continuous	age	Kept unchanged
Financial Inclusion	dig_account	Binary (Yes / No)
Digital Usage	anydigpayment, merchantpay_dig	Binary (Yes / No)
Payments	receive_wages, receive_transfers, receive_pensions, receive_agriculture	Into account / Cash / Other / Did not receive
Payments	domestic_remittances	Via account / Other / No
Payments	pay_utilities	Account / Cash / Other / Did not pay
Technology	internet_use , con1 (mobileowner)	Binary (Yes / No)

c) Cross-Wave Harmonization

After wave-specific preparation, a harmonization process was implemented to ensure consistency between 2021 and 2024. This involved standardizing variable names, aligning category structures, and simplifying residual response categories. Key variables with different naming conventions were unified, and detailed categories were collapsed into broader, analytically meaningful groups.

Table 2.3: Harmonization Actions

Type	Action
Variable Renaming	urbanicity_f2f → urbanicity
	receive_pension → receive_pensions
	remittances → domestic_remittances
	internetaccess → internet_use
	Con1 → mobileowner
Category Alignment	Remittance categories collapsed into: Via account / Other / No
Non-response Handling	Don't know / Refused / Other → grouped into No or Did not receive/pay

IV. Construction of Financial Resilience

The financial resilience indicator was constructed using the emergency funds module, specifically fin24 (availability of a source of emergency funds) and fin24a (difficulty of obtaining emergency funds within 30 days). Respondents who reported that they could not come up with emergency funds (fin24 = 7) were classified as not financially resilient, as well as those in non-response categories (don't know or refused), to ensure a conservative classification.

Among respondents reporting a potential source of emergency funds (fin24 = 1- 6), fin24a was used to assess resilience. Those reporting that it would be very difficult (1) to obtain funds were classified as not resilient, while those reporting somewhat difficult (2) or no difficulty (3) were classified as resilient. Respondents for whom fin24a was not applicable due to inability to identify any source in fin24 were also classified as not resilient.

The final variable, **fin_resilience**, is a binary indicator coded as **0 = not resilient** and **1 = resilient**. This construction ensures that both access to emergency funds and perceived ease of obtaining them are jointly captured, while avoiding sample selection bias by retaining respondents excluded from the follow-up question and classifying them consistently within the analytical framework.

V. Construction of Pre- and Post-COVID Financial Indicators

The data preparation process involved constructing indicators of payment behavior shifts for in-store transactions. Pre-COVID cash usage (fin14_2) defined baseline behavior, while post-COVID digital use was captured through mobile, debit, and credit payments (fin14_1, fin4A, fin8A). Due to skip logic, pre-COVID information was collected only for current digital users, resulting in non-digital users not being observed in the transition module. Accordingly, classification is restricted to current digital users, distinguishing between "shifted to digital" and "already digitally included", yielding a reduced analytical sample of 46 observations.

For online delivery, the analysis was limited to e-commerce users (fin14B == 1), totaling 24 observations. Pre-COVID behavior (fin14C_2) distinguished between cash-on-delivery and digital users, while post-COVID behavior (fin14C) captured current payment methods. Observations with missing pre-COVID responses were excluded due to survey design constraints to ensure comparability.

VI. Construction of Financial Exclusion Barrier Variables

To ensure comparability across waves, financial exclusion barriers were harmonized using the FIN11 module, which captures self-reported reasons for not having an account at a formal financial institution (excluding mobile money accounts). The analysis is restricted to individuals without a formal account (account_fin = "No") in both periods. A core set of barriers was consistently available across waves, including distance (too far), cost (too expensive), lack of documentation, insufficient money, lack of trust in financial institutions, and the presence of another household member with an account.

However, some differences exist between waves. In 2021, additional reasons such as religious constraints and lack of need for formal financial services were included, while these were not retained in the 2025 harmonized specification. Conversely, the 2025 wave followed a more standardized structure focusing on access, affordability, documentation, liquidity constraints, trust, and household-level substitution. These differences were addressed through harmonization to ensure that only conceptually comparable indicators were used in the pooled analysis.

Univariate Analysis



Figure 2.1: Distribution of Egypt's Population Across Income Quintiles (2021–2024)

The charts display the distribution of the population across five income quintiles, ranging from Q1 (poorest) to Q5 (richest). Comparing 2021 and 2024, the results indicate a shift toward a more balanced distribution. In 2021, a higher share of the population was concentrated in the upper income quintile, while lower quintiles accounted for a smaller proportion. By 2024, the distribution became more even, with an increase in the share of lower-income groups and a slight decline in the highest income group.

Table 2.4: Demographic and Educational Characteristics in Egypt (2021 vs 2024)

Variable	Category	2021 (%)	2024 (%)	Variable	Category	2021 (%)	2024 (%)
Gender	Female	48.6	50.9	Residence	Urban	89.0	65.0
	Male	51.4	49.1		Rural	11.0	35.0
Employment status	In workforce	45.8	45.2	Education	Primary or less	37.9	36.5
	Not in workforce	54.2	54.8		Secondary	49.6	52.2
	-	-	-		Tertiary or more	12.6	11.4

The demographic structure remains broadly stable between 2021 and 2024. Gender distribution is nearly balanced in both waves, with a slight increase in the share of females in 2024 (50.9% vs 48.6%). Labor force participation shows minimal variation. However, the share of urban residents declined in 2024 (65.0% vs 89.0%), while education levels remained largely unchanged, with only minor shifts across categories.

Table 2.5: Financial Inclusion, Behavior, and Digital & Access Indicators in Egypt (Percentage of “Yes” Responses, 2021 vs 2024)

Category	Indicator	2021 (%)	2024 (%)	Category	Indicator	2021 (%)	2024 (%)
Financial Inclusion	Account ownership	34.7	45.6	Financial Behavior	Borrowed	43.9	55.9
	Financial institution account	33.0	38.9		Saved	22.7	22.3
	Mobile money account	3.4	15.2	Digital & Access	Internet access	39.8	57.6
	Any digital payment	25.5	38.5		Mobile ownership	84.0	86.0
	Merchant digital payment	4.4	4.7		Digital account ownership	-	28.8

Financial inclusion in Egypt improved between 2021 and 2024, as account ownership increased from 34.7% to 45.6% and digital payments rose from 25.5% to 38.5%. This improvement is accompanied by a notable increase in mobile money adoption, while merchant digital payments remain low and largely unchanged. Financial behavior shows mixed patterns, with borrowing increasing from 43.9% to 55.9% and saving remaining stable at around 22% across both waves. At the same time, digital access improved substantially, as internet access rose from 39.8% to 57.6% and digital account ownership reached 28.8% in 2024, although the translation of connectivity into active financial usage remains gradual and mobile ownership remained stable (84.0% in 2021 → 86.0% in 2024).

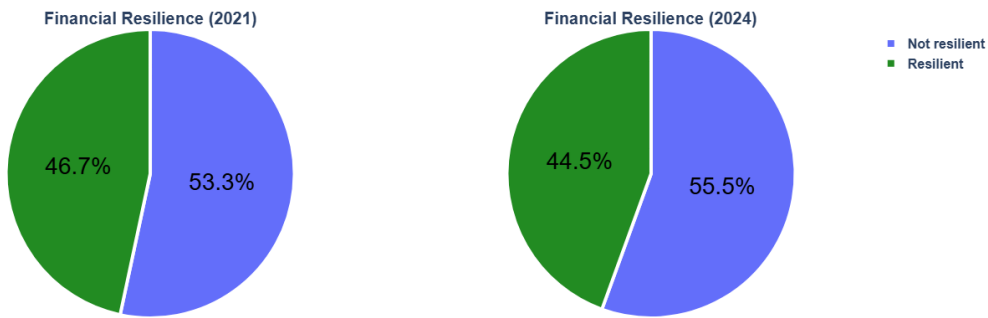


Figure 2.2: Comparison of Financial Resilience Between 2021 and 2024

The comparison between 2021 and 2024 suggests a modest decline in financial resilience. In 2021, 46.7% of respondents were financially resilient, compared to 53.3% who were not. By 2024, the share of financially resilient individuals declined to 44.5%, while non-resilient individuals increased to 55.5%, indicating a slight increase in financial vulnerability.

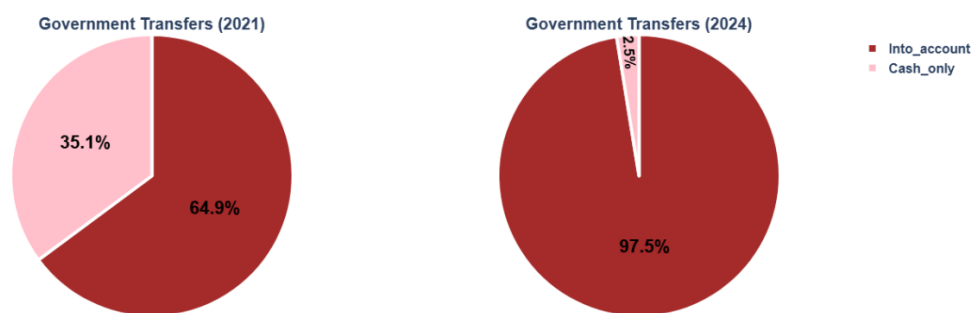


Figure 2.3: The Digital Transformation of Government Transfers in Egypt (2021–2024)

The shift from 2021 to 2024 reflects a strong transition toward digitalization in government transfer payments. In 2021, over one-third (35.1%) of transfers were still delivered in cash, which often limited efficiency in delivery. By 2024, cash payments declined sharply to 2.5%, while account-based payments increased to 97.5%, indicating near-complete digital coverage.

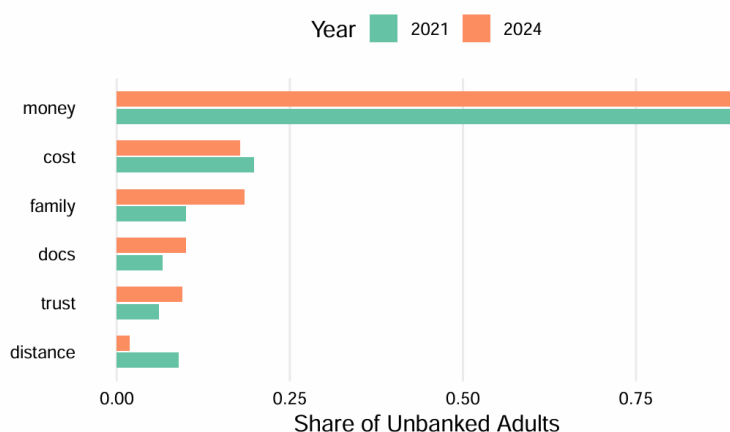


Figure 2.4: Financial Exclusion Barriers among the Unbanked in Egypt (2021 vs 2024)

The figure presents the share of unbanked adults reporting different barriers to financial inclusion in Egypt in 2021 and 2024. Insufficient money remains the dominant constraint in both years, affecting nearly 90% of unbanked individuals (89.6% in 2021 vs 88.7% in 2024). Other barriers show mixed changes over time, as lack of documentation increased from 6.6% to 10.0% and trust-related concerns rose from 6.1% to 9.5%, while the share reporting that a household member already holds an account increased from 10.0% to 18.5%. In contrast, distance to financial institutions declined sharply from 8.9% to 1.8%, while cost-related barriers remained relatively stable (19.8% vs 17.8%).

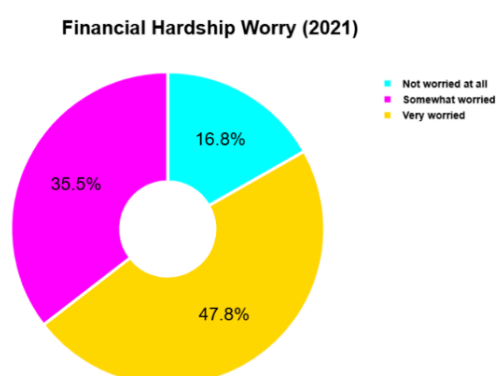


Figure 2.5: Perceived Financial Hardship Due to COVID-19 (Egypt - 2021)

The results indicate a high level of perceived financial insecurity related to the COVID-19 disruption in Egypt. Nearly half of the sample (47.8%) reported being very worried about experiencing or continuing to experience severe financial hardship, while 35.5% reported being somewhat worried. Only 16.8% stated that they were not worried at all.

Table 2.6: Payment Disruptions Across Contexts During COVID-19 (Egypt – 2021)

Payment Context	Shift Outcome	Percentage
In-store payments	Shifted to digital	71.74%
	Already digital	28.26%
Online delivery payments	Stayed cash on delivery	62.50%
	Shifted to online payment	25.00%
	Already digital	12.50%

The results indicate a clear shift toward digital payment methods in in-store payments during the COVID-19 period, where 71.74% of respondents transitioned from cash-based to digital methods, while 28.26% were already digitally included. In contrast, online delivery payments exhibit more heterogeneous behavior, as 62.5% of users continued relying on cash-on-delivery, while 25% shifted to online payment methods and 12.5% were already digitally included. Overall, the results suggest stronger digital adoption in in-store transactions compared to online delivery payments, where cash usage remains relatively persistent.

Bivariate Analysis

Table 2.7: Demographic Characteristics and Account Ownership (Egypt, 2021 vs 2024)

Variable	Category	2021 Yes (%)	2024 Yes (%)	Variable	Category	2021 Yes (%)	2024 Yes (%)
Gender	Female	30.0	40.1	Education	Primary or less	34.3	44.1
	Male	39.0	51.2		Secondary	29.4	41.8
Income Quintile	Q1 (Poorest)	23.3	34.1	Employment	Tertiary or more	56.3	67.5
	Q2	20.3	32.6		Out of workforce	31.5	41.0
	Q3	26.7	43.8		In workforce	38.4	51.1
	Q4	32.4	46.2	Residence	Rural	30.0	44.4
	Q5 (Richest)	55.4	65.8		Urban	35.2	46.2

The comparison between 2021 and 2024 shows higher account ownership across all demographic groups. Gender differences persist, with males consistently exhibiting higher levels of financial inclusion than females. Education remains a strong correlate of account ownership, as individuals with tertiary education report the highest levels in both years.

Income quintiles exhibit a clear gradient, with higher-income groups showing consistently higher account ownership compared to lower-income groups. Employment status is also positively associated with financial inclusion, as individuals in the workforce report higher ownership rates than those outside the labor force.

The urban - rural gap remains relatively small compared to socioeconomic differences, although urban residents continue to show slightly higher levels of account ownership in both periods.

Table 2.8: Financial Behavior (Saving and Borrowing) and Account Ownership (Egypt, 2021 vs 2024)

Variable	Category	2021 Yes (%)	2024 Yes (%)
Saved	No	29.1	37.3
	Yes	53.7	74.4
Borrowed	No	32.2	45.0
	Yes	37.8	46.0

The results indicate a strong association between saving behavior and account ownership in both 2021 and 2024. Individuals who report saving exhibit substantially higher levels of account ownership compared to those who do not save. Moreover, this gap widens over time, suggesting that saving behavior is closely linked to financial inclusion.

In contrast, borrowing shows only a weak association with account ownership. The proportion of individuals with accounts is relatively similar between borrowers and non-borrowers in both years, indicating that borrowing behavior does not strongly differentiate between financially included and excluded individuals. Overall, these findings highlight that financial behaviors are differentially associated with account ownership, with saving showing a much stronger relationship than borrowing.

Table 2.9: Financial Resilience and Account Ownership (2021 vs 2024)

Financial Resilience	2021 Yes (%)	2024 Yes (%)
Not resilient	26.2	40.5
Resilient	44.3	51.9

Financial resilience is positively associated with account ownership in both years, as financially resilient individuals consistently report higher levels of financial inclusion than non-resilient individuals. Account ownership increases over time for both groups, indicating overall improvement in financial inclusion between 2021 and 2024. This pattern suggests that financial inclusion is associated with greater capacity to manage financial shocks, alongside broader improvements in access to financial services over time.

Table 2.10: Internet Access, Urban–Rural Disparities, and Financial Inclusion in Egypt (2021 - 2024)

Year	Area	Internet: Yes	Internet: No	Year	Internet Access	Has Account	No Account
2021	Rural	34.5%	65.5%	2021	Yes	41.0%	59.0%
2021	Urban	40.4%	59.6%	2021	No	30.5%	69.5%
2024	Rural	50.7%	49.3%	2024	Yes	47.7%	52.3%
2024	Urban	61.2%	38.8%	2024	No	42.6%	57.4%

The results show a persistent urban-rural disparity in internet access, which widened from around 6 percentage points in 2021 (40.4% urban vs 34.5% rural) to about 10.5 percentage points in 2024 (61.2% vs 50.7%), despite overall improvements in both areas. Internet access is positively associated with financial inclusion in both years. Account ownership increases among both internet users and non-users, from 41.0% vs 30.5% in 2021 to 47.7% vs 42.6% in 2024, suggesting that financial inclusion has expanded across both connected and non-connected groups. Overall, the results indicate widening digital access disparities alongside a narrowing gap in financial inclusion by internet access status.

Chapter (3) Data Analysis

This chapter presents a comprehensive empirical investigation of financial inclusion, digital access, and financial resilience in Egypt, drawing on data from the 2021 and 2024 waves of the World Bank Global Findex Database. It adopts an integrated analytical framework that combines multiple econometric and data-driven approaches to better understand how socioeconomic characteristics shape access to digital and financial services, and how these relationships evolve over time. The central focus of the chapter is to examine financial inclusion not as an isolated outcome, but as part of a broader system linking digital connectivity, income distribution, education, and labor market participation to financial well-being.

The analysis begins by focusing on digital access, particularly internet connectivity, which is treated as a foundational requirement for participation in modern financial systems. In this context, the chapter investigates urban–rural disparities in internet access and how these gaps have changed between 2021 and 2024. It also explores the determinants of internet access using logistic regression models that incorporate key socioeconomic variables such as income quantiles, education, gender, age, employment, and urbanicity. This section highlights how digital inequality is not only persistent but also structurally shaped by underlying socioeconomic differences, with increasing divergence observed in recent years.

Building on the digital access dimension, the chapter then turns to financial inclusion and its determinants using a set of logistic regression models estimated for both years. These models allow for comparison over time and across population subgroups, and include interaction terms to capture heterogeneity in effects, particularly between gender and employment. In addition, a reverse modelling strategy is employed to examine the bidirectional relationship between financial inclusion and relative income position, measured through income quantiles. This approach provides a more nuanced understanding of whether financial inclusion is a driver of improved socioeconomic status or a reflection of it, or both.

To further understand inequality structures, the chapter applies Fairlie decomposition techniques to quantify the urban–rural gap in internet access and identify the contribution of key socioeconomic factors to this gap. This is complemented by a clustering analysis that identifies distinct financial usage profiles within the Egyptian population, revealing heterogeneous patterns of inclusion, exclusion, and digital connectivity. These clusters help to move beyond average effects and provide a more granular view of how different groups experience financial systems differently.

Finally, the chapter integrates all these dimensions using a structural equation model that traces the pathways from socioeconomic characteristics to financial inclusion, saving behaviour, and ultimately financial resilience. This framework allows for the estimation of both direct and indirect effects, particularly highlighting the role of saving as a key transmission channel through which financial inclusion translates into improved financial resilience. By combining these methods, the chapter provides a coherent and multi-layered understanding of financial inclusion dynamics in Egypt, showing how digital access, economic position, and behavioral responses jointly shape financial outcomes over time.

Digital Access and Urban–Rural Disparities

This section examines the determinants of internet access and the extent of urban–rural disparities between 2021 and 2024. Internet access is treated as a necessary condition for meaningful participation in digital financial services, making digital connectivity an important dimension of financial inclusion.

The urban–rural dimension is emphasized because spatial inequalities often reflect differences in digital infrastructure, service availability, and accessibility. Comparing the two periods allows the analysis to capture not only the existence of disparities, but also how the structure of digital inequality evolved over time.

I. Variable Construction and Recoding

To ensure sufficient cell sizes and improve the stability and interpretability of the estimated models, categorical variables were recoded into broader and economically meaningful groups. Income quintiles were aggregated into a binary indicator distinguishing the **lower 60% (Q1–Q3)** from the **upper 40% (Q4–Q5)**. This classification is motivated by capturing a clear socioeconomic divide between lower- and middle-income populations versus relatively higher-income (more affluent) groups. This distinction is particularly relevant for studying digital access, as financial constraints and affordability of internet services are expected to differ substantially between these segments.

II. Model Specification

The analysis examines the determinants of internet access using a set of socioeconomic characteristics, including income group, gender, education level, age, employment status, and urbanicity. To capture both temporal and spatial differences, separate models were estimated for 2021 and 2024. In addition, subgroup models for urban and rural populations were estimated as inputs for the Fairlie decomposition analysis. This results in four subgroup specifications:

- Urban 2021
 - Rural 2021
- Urban 2024
 - Rural 2024

These models provide the basis for examining both the determinants of internet access and the sources of urban–rural disparities over time.

III. Multicollinearity Diagnostics

Table 3.1: Multicollinearity Diagnostics for Urban and Rural Logistic Regression Models in Egypt (2021–2024)

Variable	Urban Model (2021) Adjusted VIF	Rural Model (2021) Adjusted VIF	Urban Model (2024) Adjusted VIF	Rural Model (2024) Adjusted VIF
Income Group	1.048	1.133	1.062	1.020
Gender	1.157	1.302	1.158	1.354
Education	1.044	1.095	1.026	1.013
Age	1.099	1.091	1.089	1.028
Employment	1.169	1.290	1.167	1.340

The VIF results indicate that multicollinearity is not a concern in any of the estimated models. All adjusted VIF values are very close to 1, which indicates almost no correlation among the explanatory variables. None of the variables approach even the conservative threshold of 5, beyond which multicollinearity may become problematic. The results are consistent across both years (2021 and 2024) and both subsamples (urban and rural). Although slightly higher values are observed for gender and employment in the rural 2024 model, these remain well within acceptable limits and do not pose any threat to model stability.

IV. Logistic Regression

To examine the determinants of internet access, a binary logistic regression model is employed. The dependent variable is a binary variable equal to one if the individual has access to the internet and zero otherwise. Given the binary nature of the outcome, logistic regression is appropriate as it models the probability of access as a nonlinear function bounded between zero and one. The model is estimated separately for 2021 and 2024 using survey - weighted logistic regression to account for the sampling design and ensure population-representative estimates. The specification includes key socio-economic characteristics such as income group, gender, education level, age, employment status, and urban–rural location.

$$p(\text{Internet} = 1) = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 \cdot \text{Income} + \beta_2 \cdot \text{Gender} + \beta_3 \cdot \text{Education} + \beta_4 \cdot \text{Age} + \beta_5 \cdot \text{Workforce} + \beta_6 \cdot \text{Urbanicity})]} \quad (3.1)$$

Estimating separate models for each year allows for differences in the effects of these variables over time, thereby capturing potential structural changes in the determinants of internet access. The estimated models are subsequently used as inputs in the decomposition analysis to quantify the contribution of observed characteristics to urban–rural disparities.

V. Results for 2021's Model:

Table 3.2: Logistic Regression Results for Determinants of Internet Access in Egypt (2021)

Coefficients	Odds Ratio	95% Confidence Interval	P-value
Intercept	0.4862190	(0.1824451 , 1.2957809)	0.149
Income	1.5397143	(1.0534499 , 2.2504347)	0.026
Gender	0.7114443	(0.4640565 , 1.0907140)	0.118
Education (Secondary)	4.3004926	(2.7329216 , 6.7672035)	0.000
Education (Tertiary or more)	27.427745	(13.4679177 , 55.8572778)	0.000
Age	0.9739118	(0.9604679 , 0.9875439)	0.000
Employment	1.1496882	(0.7396582 , 1.7870187)	0.535
Urbanicity	1.0260312	(0.5498735 , 1.9145131)	0.936

In 2021, internet access in Egypt was mainly driven by socioeconomic factors, with education being the strongest determinant. Higher-income individuals were about 54% more likely to have internet access than lower-income groups. Education had a strong non-linear effect, where secondary education increased the odds by about 4.3 times, and tertiary education by more than 27 times compared to primary education or less. Age had a significant negative effect, with older individuals less likely to have internet access. In contrast, gender, employment status, and urbanicity were not statistically significant after controlling for other factors.

VI. Results of 2024 Model

Table 3.3: Logistic Regression Results for Determinants of Internet Access in Egypt (2024)

Coefficients	Odds Ratio	95% Confidence Interval	P-value
Intercept	2.665985	(1.1575910 , 6.1398835)	0.021
Income	2.637135	(1.7467953 , 3.9812801)	0.000
Gender	0.493742	(0.3123474 , 0.7804808)	0.003
Education (Secondary)	3.835382	(2.5899843 , 5.6796321)	0.000
Education (Tertiary or more)	23.071041	(9.9292659 , 53.6064721)	0.000
Age	0.942286	(0.9283499 , 0.9564313)	0.000
Employment	1.562245	(0.9856261 , 2.4762017)	0.058
Urbanicity	2.028263	(1.3739544 , 2.9941679)	0.000

By 2024, the determinants of internet access became stronger and more clearly differentiated, indicating a sharper pattern of digital inequality. Income remained highly significant, with individuals in the upper 40% income group about 164% more likely to have internet access compared to lower-income groups. Education continued to be a dominant factor, where secondary education increased the odds by about 3.8 times and tertiary education by more than 23 times.

Gender became statistically significant, with females about 51% less likely to have internet access than males, indicating an emerging gender digital divide. Age had a stronger negative effect, reflecting a widening generational gap. Employment became marginally significant, **while urbanicity became strongly significant, with urban residents about twice as likely to have internet access compared to rural residents, highlighting a clear spatial divide.**

VII. Models Performance

Year	Accuracy	Area under ROC Curve
2021	73.2%	81.2%
2024	77.6%	84.6%

The model performance improves from 2021 to 2024, indicating better predictive power in explaining internet access over time. Classification accuracy increases from 73.2% in 2021 to 77.6% in 2024, suggesting that the explanatory variables capture the determinants of internet access more effectively in the later period, although accuracy should be interpreted alongside AUC due to its dependence on the classification threshold. The Area Under the ROC Curve (AUC) also rises from 0.812 to 0.846, confirming strong and improving discriminative ability in both years. This means the model correctly ranks an internet user higher than a non-user about 81% of the time in 2021, increasing to about 85% in 2024. Overall, both ROC curves lie well above the 45-degree line, with the 2024 curve consistently outperforming 2021, indicating superior model performance across all thresholds.

VIII. Fairlie Decomposition

Fairlie decomposition results were used to examine the sources of urban–rural disparities in internet access in 2021 and 2024, separating the explained component driven by observable socioeconomic characteristics from the unexplained structural component. The population structure shifted notably over time, with the urban share decreasing from about 89% in 2021 to 64% in 2024, while the rural share increased from 11% to 36%. At the same time, overall internet access rose from 37.7% to 54.9%. These changes indicate that improvements in internet access are influenced not only by changes in individual characteristics but also by shifts in population composition, particularly the increased representation of rural populations in 2024, which may have contributed to a more pronounced urban–rural digital divide.

Table 3.4: Fairlie Decomposition of Urban–Rural Gap in Internet Access (2021 vs 2024)

Year	Total Gap	Explained	Unexplained	Explained (Urban Reference)	Explained (Rural Reference)
2021	0.002143897	0.01013752	-0.007993625	0.01754946	0.00272558
2024	0.1510194	0.1044086	0.04661072	0.1019868	0.1068304

In 2021, the urban–rural gap in internet access was extremely small (0.2 percentage points). The explained component (0.0101) was larger than the total gap, indicating that observable characteristics such as income, education, and employment would predict higher urban access, but this did not translate into an actual gap. The unexplained part was negative (–0.0080), suggesting offsetting factors that reduced inequality, while results varied slightly across reference groups due to the very small overall gap.

In 2024, the gap widened significantly to 15.1 percentage points, indicating a clear urban advantage. Most of the gap (about 69%) is explained by differences in characteristics such as income, education, and employment (0.1044), while the remaining unexplained portion (0.0466) suggests structural advantages in urban areas. The results are stable across reference groups, confirming a robust and persistent urban–rural digital divide.

IX. Variable Contributions for Digital Divide

To further examine the explained component of the Fairlie decomposition, variable-level contributions were estimated to identify which socioeconomic factors contributed most to the urban–rural gap in internet access. The contribution estimates were computed using a symmetric Fairlie decomposition procedure with repeated resampling to ensure robustness and reference-group invariance. Detailed methodological procedures are provided in the Methodology chapter.

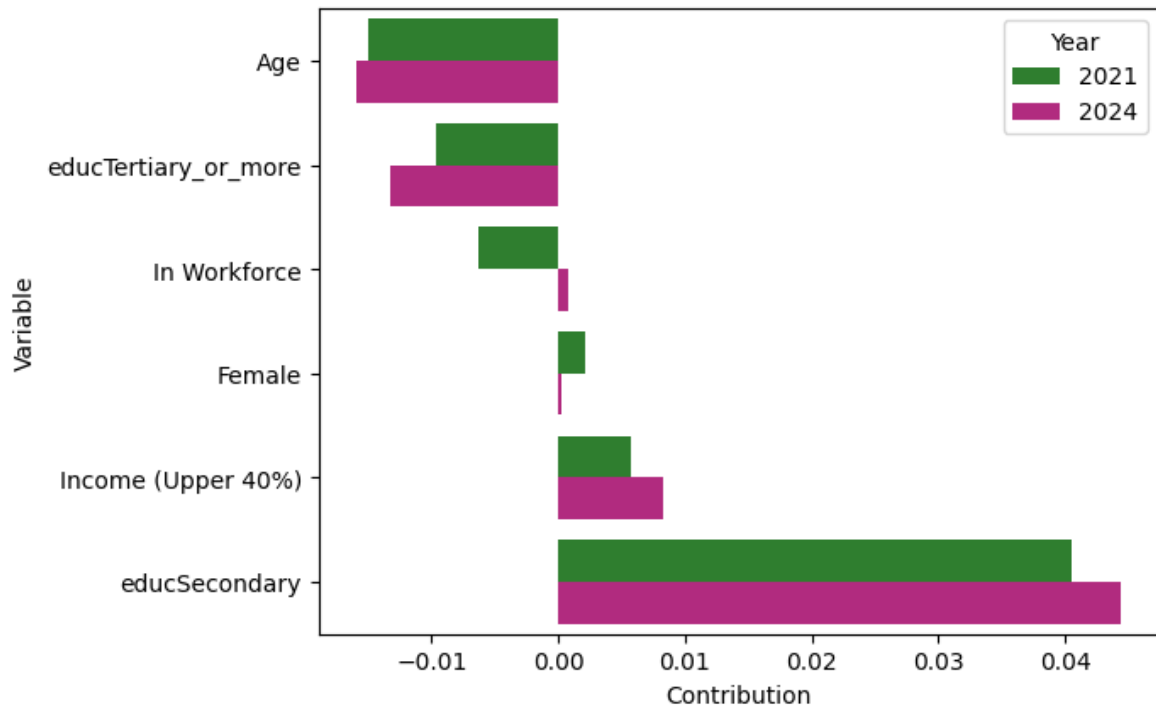


Figure 3.1: Decomposition of Urban–Rural Internet Access Gap by Socioeconomic Variables (2021–2024)

The comparative analysis reveals a clear structural shift in the nature of inequality in internet access between 2021 and 2024. While the earlier period is characterized by a near absence of disparity, the later period shows the emergence of a well-defined and economically meaningful digital divide.

This evolution occurs alongside a notable change in sample composition, where the share of rural respondents increases from 11.0% in 2021 to 35.0% in 2024, while the urban share declines from 89.0% to 65.0%. This broader rural representation in 2024 is important for interpretation, as it increases heterogeneity and may partly amplify observed disparities in access.

In 2021, the urban–rural gap in internet access is minimal, and the decomposition results show relatively small and offsetting variable contributions. Secondary education is the main factor widening the gap, with a contribution of +0.0405, indicating that urban areas have a slightly higher concentration of individuals with this education level. Income also contributes positively, though marginally (+0.0057), followed by gender (+0.0021), suggesting weak demographic advantages in urban areas.

In contrast, several variables work to reduce inequality. Age contributes -0.0150 , reflecting lower internet usage among older individuals in both rural and urban areas. Tertiary education also narrows the gap (-0.0097), while employment has a small equalizing effect (-0.0062). The overall pattern in 2021 is therefore characterized by very small, partially cancelling effects, consistent with a weak and non-structural digital divide.

By 2024, the contribution pattern becomes stronger and more structured, reflecting a more pronounced urban–rural divide. Secondary education remains the dominant driver of inequality, increasing the gap by +0.0444, highlighting the growing importance of intermediate human capital in determining internet access. Income also becomes more relevant, contributing (+0.0083), followed by employment (+0.0008) and gender (+0.0003), all reinforcing urban advantage, albeit modestly.

On the other hand, age continues to reduce the gap significantly (−0.0159), indicating a consistent age-related digital usage pattern across both regions. Interestingly, tertiary education also reduces inequality in 2024 (−0.0133), possibly reflecting either saturation effects among highly educated individuals or the presence of highly educated individuals outside urban centers.

Overall, the comparison shows that while 2021 is characterized by weak and largely offsetting effects across variables, 2024 exhibits stronger and more systematic contributions, particularly from education and income. This shift reflects a transition from a diffuse pattern of access to a more structured and economically stratified digital divide.

Financial Usage Profiles in the Egyptian Population

Clustering is a useful method to group observations based on their similarities. It helps simplify complex data and identify clear patterns between individuals. In this analysis, clustering is used to examine differences in financial inclusion, digital access, and socioeconomic characteristics, in order to classify individuals into meaningful groups.

I. Model Construction

The analysis was conducted using the **K-Modes clustering algorithm**, which is the standard method for grouping individuals based on categorical data. This approach allows for the identification of clear patterns and the division of the sample into distinct groups (clusters) that share similar characteristics. To ensure robust and meaningful results, the clusters were constructed based on **five core variables**:

- **Financial Account Ownership (account_fin):** To measure formal banking access.
- **Digital Payment Usage (anydigpayment):** To measure the use of modern financial technology.
- **Internet Access (internetaccess):** To assess digital connectivity.
- **Educational Level (educ):** To reflect the educational background of individuals.
- **Employment Status (emp_in):** To represent their integration into the labor market.

Following the formation of the main clusters, a **profiling analysis** was performed. In this stage, the distribution of additional key factors such as **Gender**, **Urbanicity**, and **Financial Resilience** was examined within each cluster. This process provides a deeper understanding of the social and economic identity of each group without influencing the initial clustering logic.

II. Analysis for 2021

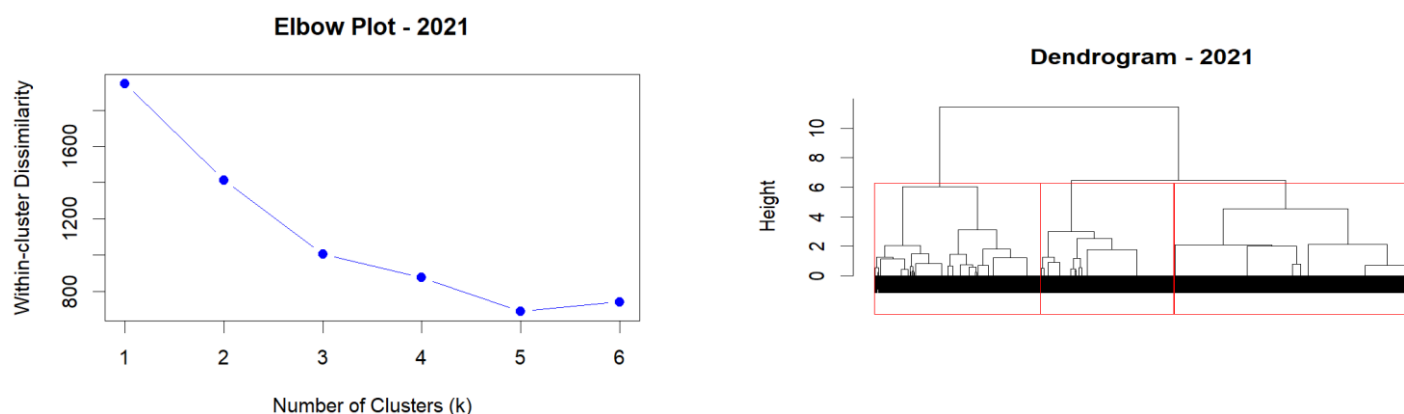


Figure 3.2: Elbow Plot and Dendrogram – 2021

The Elbow plot indicates a gradual leveling off around 3 to 5 clusters. Although several solutions are plausible, the three-cluster specification is considered the most balanced and interpretable. This choice is further supported by the dendrogram, which clearly reveals three distinct groups, confirming that the data can be appropriately segmented into three clusters.

Table 3.5: 2021 Cluster Profiles

Indicator (%)	Structurally Excluded	Digitally Connected Excluded	Fully Included
Sample Size (N)	453	329	219
Female Rate	57.0	48.0	32.0
Internet Access	3.53	72.3	65.8
Financial Account	23.4	5.47	94.1
Digital Payments	13.9	1.22	85.8
Financial Resilience	36.4	45.3	69.9
Primary Educ. or Less	73.5	5.47	12.8

The clustering results reveal three distinct financial and digital inclusion profiles within the Egyptian population in 2021, reflecting substantial heterogeneity in access to financial services, digital connectivity, and socioeconomic characteristics.

- Cluster 1 (Structurally Excluded):** This segment represents the largest portion of the sample, characterized by severe digital poverty (3.53% internet access) and low educational capital (73.5% primary education). The group is predominantly female and faces systemic barriers to both digital and financial ecosystems.
- Cluster 2 (Digitally Connected Excluded):** A paradox is observed here; despite high connectivity (72.3%) and better education, financial inclusion remains critically low (5.47% account ownership). This suggests that digital access does not automatically translate into financial integration for this group.
- Cluster 3 (Fully Included):** This cluster reflects the demographic "elite" in 2021 - mostly male, highly resilient (69.9%), and effectively utilizing both banking and digital payment systems.

III. Analysis for 2024

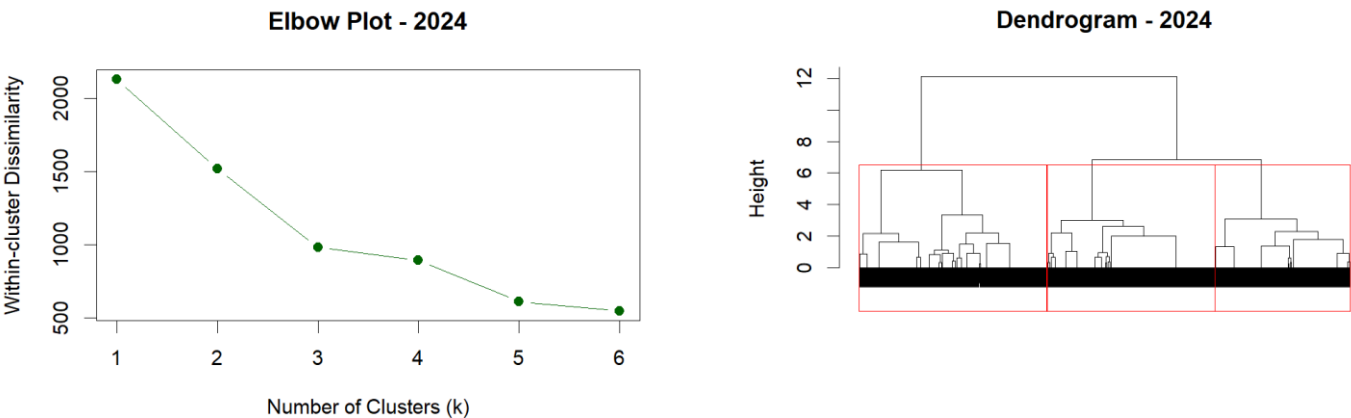


Figure 3.3: Elbow Plot and Dendrogram – 2024

The Elbow plot displays a similar pattern, with a clear leveling off between 3 and 5 clusters. The three-cluster solution is considered the most balanced and interpretable, and this is further confirmed by the dendrogram. The dendrogram reveals three distinct clusters, supporting the robustness of this clustering structure.

Table 3.6: 2024 Cluster Profiles

Indicator (%)	Fully Included	Structurally Excluded	Digitally Connected Excluded
Sample Size (N)	296	339	366
Female Rate	34.1	63.1	53.0
Internet Access	78.4	7.96	86.6
Financial Account	89.2	32.4	4.10
Digital Payments	92.6	24.8	7.38
Financial Resilience	55.4	27.7	51.1
Primary Educ. or Less	15.2	83.8	9.84

The clustering results reveal three distinct financial and digital inclusion profiles within the Egyptian population in 2024, similar to 2021, reflecting substantial heterogeneity in access to financial services, digital connectivity, and socioeconomic characteristics.

- 1) **Cluster 1 (Fully Included):** This group has matured, with digital payment usage rising to 92.6%. However, their financial resilience dropped compared to 2021 (from 69.9% to 55.4%), likely reflecting broader macroeconomic pressures.
- 2) **Cluster 2 (Structurally Excluded):** The gender gap has widened significantly, with females now constituting 63.1% of this vulnerable group. Despite a slight increase in account ownership (32.4%), their lack of education (83.8% primary) and internet access remains a barrier to meaningful inclusion.
- 3) **Cluster 3 (Digitally Connected Excluded):** Now the largest cluster (N=366), this group shows extremely high internet penetration (86.6%) but continues to be alienated from the formal banking sector (4.1% accounts). They represent a significant untapped market for fintech interventions

IV. Evolution of Financial and Digital Inclusion Clusters in Egypt (2021–2024)

The clustering results for 2021 and 2024 show a stable three-group structure, with important shifts in the composition and intensity of financial and digital inclusion profiles over time. In 2021, the system is relatively less differentiated, with a large Structurally Excluded group characterized by very low internet access (3.53%) and low education levels (73.5% primary or less). The Digitally Connected Excluded cluster exhibits high internet access (72.3%) but very low financial inclusion (5.47%), indicating a clear disconnect between digital connectivity and formal financial participation. The Fully Included group remains small but highly integrated, with strong account ownership (94.1%) and digital payment usage (85.8%).

By 2024, the clustering structure becomes more polarized, with sharper differences across groups. The Fully Included cluster shows higher digital and financial engagement, with account ownership at 89.2% and digital payments at 92.6%, despite a decline in financial resilience. The Structurally Excluded group remains highly disadvantaged, with persistently low internet access (7.96%) and high prevalence of low education (83.8%), alongside increased female representation. The Digitally Connected Excluded group expands to become the largest cluster (N = 366), combining very high internet access (86.6%) with extremely low financial account ownership (4.10%), reinforcing a persistent gap between digital access and financial inclusion.

Financial Inclusion, Relative Poverty, and Their Determinants

The analysis employs direct models to evaluate financial inclusion using data from the World Bank. The two cross-sectional surveys from 2021 and 2024 are pooled into a single dataset, with a year dummy included to distinguish between the two periods and capture temporal differences in financial inclusion and its determinants. This pooled structure allows more efficient estimation while enabling comparison of structural changes over time within a unified framework.

Financial inclusion is measured using a binary indicator of account ownership (account), which equals 1 if the respondent has an account at a financial institution, a mobile money account, or both, and 0 otherwise. This captures both traditional and digital forms of financial access.

The model is estimated using logistic regression with survey weights to ensure population representativeness. Key explanatory variables include internet access, age, education, employment status, and income quantiles (as a proxy for relative poverty), used to identify the main determinants of financial inclusion. This specification reflects individuals' position within the income distribution rather than absolute income levels, allowing assessment of relative socioeconomic disadvantage in explaining financial access.

In addition, this section examines the relationship between income quantiles and account ownership through an interaction model and a reverse model approach. Income quantiles are grouped into two categories: lower group (Q1–Q3) and upper group (Q4–Q5), to facilitate relative poverty analysis. This framework accounts for the potential two-way relationship between the variables, allowing assessment of how their association evolves over time rather than assuming a single-direction relationship.

I. The Initial Access Model

The Initial Access Model investigates the determinants of financial inclusion in Egypt. It focuses on income quantiles, employment status, and internet access, while controlling for key demographic characteristics. A logistic regression framework is used to estimate the probability of financial inclusion, defined as holding either a financial institution or mobile money account. Income quantiles, age, and education are included as main explanatory variables. The econometric specification of the Initial Access Model is defined as follows:

$$\text{logit}(P(\text{Account} = 1)) = \beta_0 + \beta_1 \text{UpperIncomeQuantile} + \beta_2 \text{Year} + \beta_3 \text{Age} + \beta_4 \text{Internet} + \beta_5 \text{Gender} + \beta_6 \text{Education} + \beta_7 \text{Employment} + \beta_8 \text{Urbanicity} + \varepsilon \quad (3.2)$$

(A) Assumptions

- **Multicollinearity:** VIF values were checked to test multicollinearity. All results are below the threshold. Income is 1.078, education is 1.164, and year is 1.385. Other variables also showed low values, including age (1.563), internet access (1.515), and employment (1.549). Gender was 1.183 and urbanicity was 1.135. These numbers show there is no overlap between predictors, meaning the coefficients are stable and reliable.
- **Sparse Categories:** The cross-tabs were checked for any sparse categories that could affect the model. Since all categories have enough observations and there are no empty cells, the model is expected to converge properly and provide reliable results.
- **Check Linearity for Age:** The Box-Tidwell test was used to check if Age is linear, instead of just assuming it. The p-value came out to 0.508. This confirms a linear relationship with financial inclusion, so there was no need to change the Age variable.
- **Investigating Education Categories**

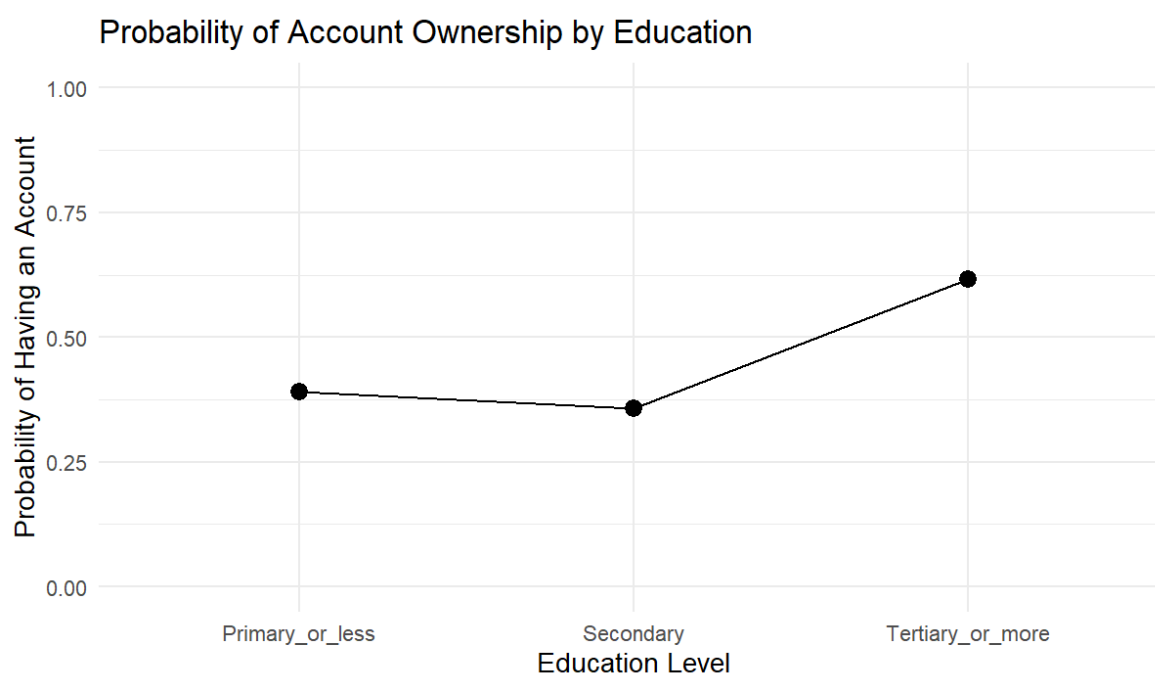


Figure 3.4: Probability of Account Ownership by Education Level

Education levels were analyzed, and it was observed that the probability of financial inclusion is similar across lower education groups. The probability for 'Primary or less' is 0.39, which is close to 0.36 for 'Secondary' education. Given this similarity in patterns, the two categories were merged into a single group, 'Secondary or less', to improve model stability. This specification provides a clearer contrast with the 'Tertiary or more' group, where the probability increases to over 0.60.

(B) Model Performance and Interpretations

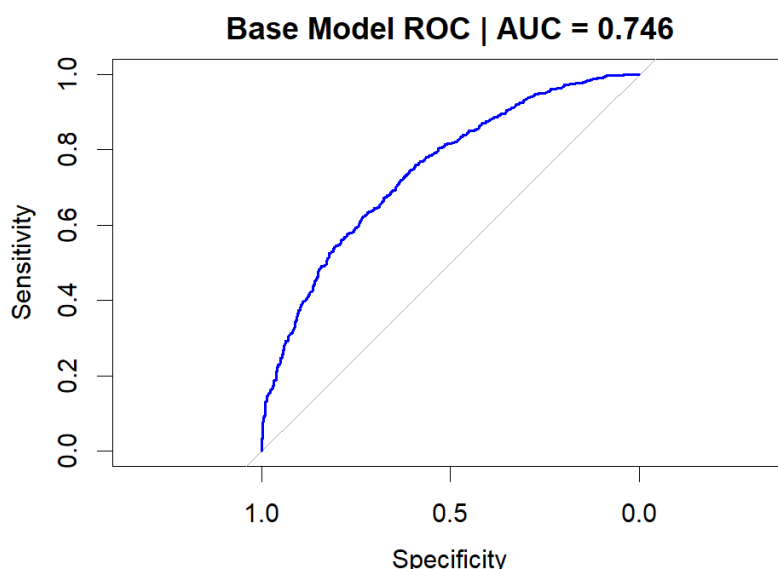


Figure 3.5: The ROC Curve of The Initial Access Model

The AUC is 0.746, indicating that the model has a good ability to distinguish between financially included and excluded individuals. Overall accuracy is approximately 70%, with a sensitivity of 62% and a specificity of 73%. The Youden Index was used to determine the optimal cutoff point, which was 0.41. These results suggest that the model performs slightly better in identifying financially excluded individuals, while still maintaining satisfactory overall predictive performance.

Table 3.7: Odds Ratio of The Initial Access Model

Variable	OR	CI Lower	CI Upper	P-value
Intercept	0.049	0.029	0.081	0.001
Upper Income Quantile Group	1.468	1.152	1.870	0.002
Year 2024	2.096	1.620	2.713	0.001
Age	1.048	1.039	1.058	0.001
Internet Access (No)	0.573	0.426	0.770	0.001
Gender (Male)	1.268	0.946	1.701	0.112
Education (Tertiary or More)	2.775	1.954	3.941	0.001
Employment (In workforce)	1.115	0.830	1.497	0.469
Urbanicity (Urban)	1.036	0.773	1.388	0.815

- **The Significant Factors:** The results highlight that Upper Income Quantile Group (Q4–Q5) is positively associated with financial inclusion (OR = 1.47), indicating that individuals in higher income positions are more likely to be financially included compared to those in lower quantiles (Q1–Q3). Year 2024 shows a strong positive effect (OR = 2.10), reflecting a substantial increase in financial inclusion relative to 2021. This is consistent with the ongoing expansion of the "Digital Egypt" agenda. Tertiary education remains the strongest determinant (OR = 2.77), nearly tripling the likelihood of financial inclusion compared to lower education levels. While income position and age contribute positively, lack of internet access (OR = 0.57) remains a major constraint, reducing the odds by about 43%, highlighting the persistence of the digital divide.
- **The Insignificant Factors:** Gender and urbanicity are not statistically significant ($p > 0.05$), indicating that financial inclusion is no longer strongly differentiated by gender or place of residence once other factors are controlled for. Employment also does not show a statistically significant standalone effect. Overall, the results suggest a gradual structural shift in financial inclusion patterns, where education and digital access are becoming more important determinants than demographic characteristics.

II. The Conditional Effects Model

The analysis focuses on changes between 2021 and 2024, with particular attention to the interaction between income quantiles and year to assess whether the effect of income position has changed over time. It also examines the interaction between gender and employment to determine whether the effect of employment on financial inclusion differs between men and women. These interaction terms capture potential shifts in the structure of financial inclusion and the extent to which access has become more equitable over time. The econometric specification of the Conditional Effects Model is defined as follows:

$$\text{logit}(P(\text{Account} = 1)) = \beta_0 + \beta_1 \text{Upper Income Quantilegroup} + \beta_2 \text{Year} + \beta_3 (\text{Upper Income Quantile} \times \text{Year}) + \beta_4 \text{Gender} + \beta_5 \text{Employment} + \beta_6 (\text{Gender} \times \text{Employment}) + \beta_7 \text{Age} + \beta_8 \text{InternetAccess} + \beta_9 \text{Education} + \beta_{10} \text{Urbanicity} + \varepsilon \quad (3.3)$$

Model Performance and Interpretations

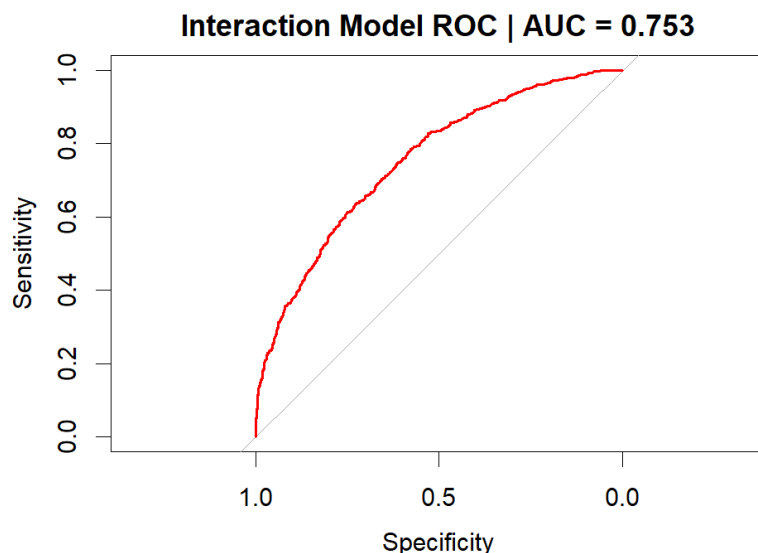


Figure 3.6: The ROC Curve of The Conditional Effects Model

The interaction model achieved an AUC of 0.753, indicating improved overall performance. Specificity reached 75%, while sensitivity was 61.4%, with overall accuracy of approximately 71%. The Youden Index was used to determine the optimal cut-off point, which was 0.41. These results indicate that the model performs better in identifying individuals outside the financial system. The inclusion of the gender - employment interaction term improves model fit by capturing heterogeneity in financial inclusion patterns across groups.

Table 3.8: Odds Ratio of The Conditional Effects Model

Variable	OR	CI Lower	CI Upper	P-value
Intercept	0.044	0.026	0.075	<0.001
Upper Income Quantile Group	1.395	0.976	1.993	0.068
Year 2024	1.989	1.382	2.863	<0.001
Gender (Male)	2.154	1.485	3.125	<0.001
Employment (In workforce)	2.253	1.479	3.432	<0.001
Age	1.048	1.039	1.058	<0.001
Internet Access (No)	0.574	0.427	0.773	<0.001
Education (Tertiary or More)	2.920	2.048	4.162	<0.001
Urbanicity (Urban)	1.009	0.751	1.356	0.953
Upper Income Quantile × Year	1.135	0.702	1.837	0.605
Gender × Employment	0.290	0.166	0.506	<0.001

- The Significant Factors:** The inclusion of interaction terms provides a more nuanced understanding of the relationship between gender, employment, and financial inclusion in Egypt. The most notable result is the interaction between gender and employment (OR = 0.29, $p < 0.001$). Since the coefficient is significantly below one, this indicates that the positive effect of employment on financial inclusion is substantially stronger for women than for men. For men, employment has a relatively smaller marginal effect, as financial inclusion is already higher among employed males. In contrast, for women, employment plays a more critical role in facilitating access to the financial system, potentially due to institutional mechanisms such as payroll-linked accounts or increased financial autonomy associated with formal employment.
- The Insignificant Factors:** Some effects are not statistically significant. For example, the interaction between Upper Income Quantile Group and year 2024 (OR = 1.135, $p = 0.605$) is not statistically meaningful, suggesting that the relationship between income position and financial inclusion has not changed significantly over time. This implies that recent improvements in digital financial inclusion have not substantially altered existing disparities across income groups.

III. Relative Poverty Determinants (Reverse Model)

The Relative Poverty Determinants Model is designed to examine the bidirectional relationship between financial inclusion and socioeconomic status. It explores whether financial inclusion is associated with a higher likelihood of belonging to upper income quantiles, while also allowing for the possibility that income position is correlated with access to financial services. This framework enables the assessment of a dual relationship between the two dimensions within a unified econometric specification. The econometric specification of the Relative Poverty Determinants Model is defined as follows:

$$\text{logit}(P(\text{UpperIncomeQuantileGroup} = 1)) = \beta_0 + \beta_1 \text{Account} + \beta_2 \text{Year} + \beta_3 \text{ns}(\text{Age}, 3) + \beta_4 \text{InternetAccess} + \beta_5 \text{Gender} + \beta_6 \text{Education} + \beta_7 \text{Employment} + \beta_8 \text{Urbanicity} + \varepsilon \quad (3.4)$$

(A) Assumptions

- **Multicollinearity:** Looking at the revised VIF results, the model seems perfectly fine. Even with the age splines, the adjusted GVIF values are low enough that multicollinearity isn't an issue at all. For instance, account and year are basically at the baseline (1.076 and 1.073), while urbanicity is even lower at 1.047. Even the variables that usually overlap, like internet access (1.249), employment (1.250), and gender (1.241), stayed well within range. Along with educ at 1.191 and age at 1.138, everything is stable. VIF results show no multicollinearity issues. All predictors are within the acceptable range, meaning the coefficients are stable for the final analysis.
- **Check Linearity for Age:** The Box-Tidwell test indicated a p-value of 0.0018, confirming a non-linear relationship between age and income. This violates the linearity assumption required in standard logistic regression models. To address this issue, spline functions (restricted cubic splines) were introduced for the age variable, allowing for a flexible functional form that captures non-linear patterns in the data. This approach ensures that the model reflects the actual distributional relationship between age and income in Egypt, rather than imposing a restrictive linear specification.
- **Investigating Education Categories**

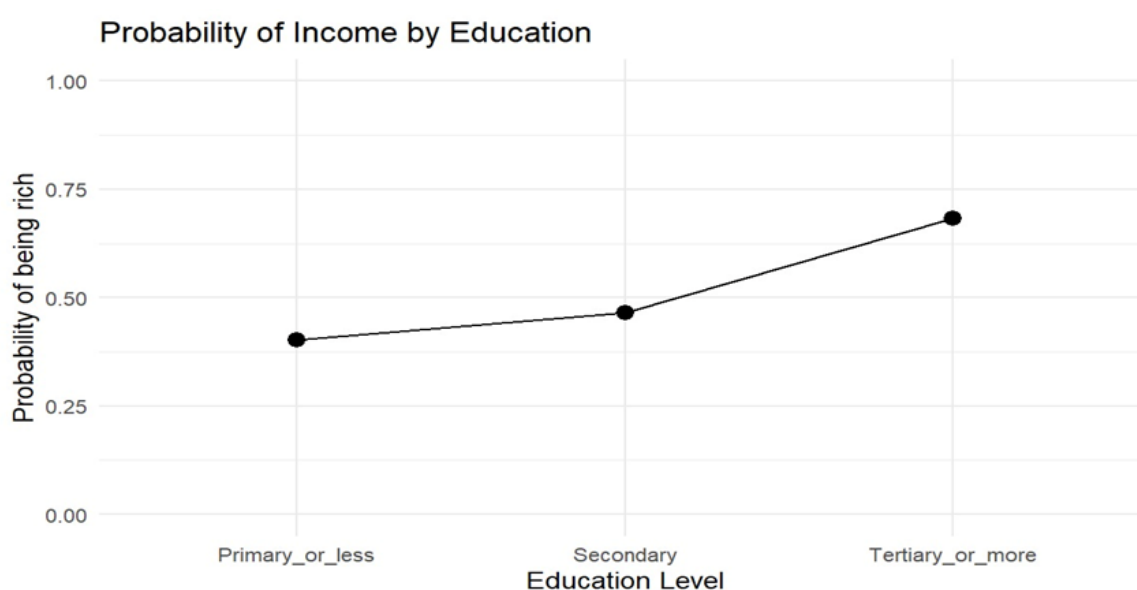


Figure 3.7: Probability of Being in the Upper Income Quantile Group by Education Level

The relationship exhibits a clear monotonic pattern the probability of belonging to higher income quantiles starts at around 0.40 for the lowest education group, increases slightly to approximately 0.47 for secondary education, and then rises more substantially to nearly 0.68 for tertiary education. This stepwise structure justifies maintaining three distinct education categories, as it preserves meaningful heterogeneity across groups and avoids oversimplification that could mask important differences in relative poverty distribution. Overall, this specification improves interpretability while maintaining the economic gradient in the model.

(B) Model Performance and Interpretations

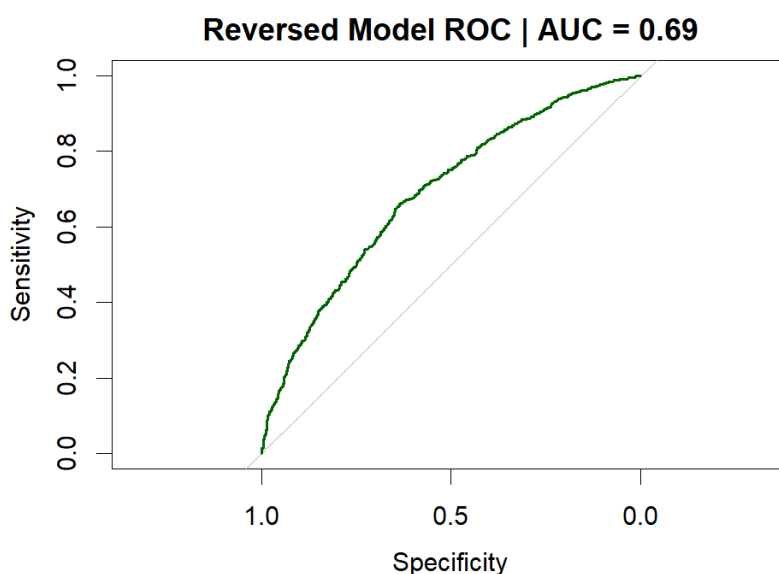


Figure 3.8: The ROC Curve of the Relative Poverty Determinants Model

Model accuracy is 66.7%, with an AUC of 0.69, indicating a moderate ability of the model to distinguish between individuals in the upper income quantile group (Q4–Q5) and the lower income quantile group (Q1–Q3). The Youden Index identifies an optimal cut-off point of 0.42. Sensitivity is 65.96%, while specificity is 63.66%. Overall, the model shows balanced predictive performance in classifying individuals across the two constructed income quantile groups.

Table 3.9: Odds Ratio of Relative Poverty Determinants Model

Variable	OR	CI Lower	CI Upper	P-value
Intercept	0.392	0.225	0.683	<0.001
Account (Yes)	1.394	1.093	1.778	0.008
Year 2024	0.863	0.679	1.098	0.230
Age (Spline 1)	2.070	1.051	4.076	0.035
Age (Spline 2)	6.516	2.076	20.455	0.001
Age (Spline 3)	12.337	3.196	47.625	<0.001
Internet Access (No)	0.515	0.390	0.678	<0.001
Gender (Male)	1.231	0.936	1.620	0.137
Secondary Education	1.621	1.212	2.168	0.001
Education (Tertiary or More)	3.879	2.501	6.017	<0.001
Employment (In workforce)	0.778	0.588	1.031	0.081
Urbanicity (Urban)	1.107	0.834	1.471	0.480

- **The Significant Factors:** The logic is reversed to examine the determinants of belonging to the upper income quantiles in Egypt. Financial inclusion is positively associated with income position (OR = 1.394), increasing the odds of being in the top income quantiles by approximately 39% compared to those without accounts. This suggests that financial inclusion is closely linked to improved socioeconomic positioning. Education remains the strongest determinant, as having a tertiary degree (OR = 3.879) increases the likelihood of being in the upper income quantiles by nearly four times relative to lower education levels. Secondary education is also significant (OR = 1.621), indicating a clear educational gradient. Age has a strong positive effect across all spline terms (Age spline 1–3), reflecting a non-linear relationship between age and income position.
- **The Insignificant Factors:** Some expected determinants are statistically insignificant. Urbanicity (OR = 1.107) shows no significant impact. Gender (OR = 1.231, $p = 0.137$) is also not statistically significant, indicating that income position is not systematically differentiated by gender once other factors are controlled for. Employment status is marginally insignificant (OR = 0.778, $p = 0.081$), while year (OR = 0.863, $p = 0.230$) shows no significant temporal shift. Overall, the results suggest that differences in income position are increasingly driven by education and financial access rather than geographic location or gender.

IV. Summary

This section analyzes financial inclusion in Egypt using pooled data from 2021 and 2024 through direct, interaction, and reverse econometric models, capturing both determinants and bidirectional relationships with relative income position.

The initial access model shows that financial inclusion is significantly driven by income position, education, age, internet access, and time effects. Higher income quantiles (Q4–Q5) and tertiary education strongly increase the likelihood of financial inclusion, while internet access remains a key constraint. The year effect is positive and significant, indicating a clear increase in financial inclusion in 2024 compared to 2021, reflecting improvements in digital financial access over time.

However, this time effect does not carry over to the reverse model, where the year variable becomes insignificant, suggesting that while access to financial services has improved, the underlying distribution of income positions has not changed significantly. The conditional effects model further highlights important heterogeneity, where the interaction between gender and employment is significant, indicating that employment has a stronger impact on financial inclusion for women than for men.

Finally, the reverse model confirms a bidirectional relationship between financial inclusion and income position. Education remains the strongest determinant of being in higher income quantiles, while age shows a non-linear relationship with income but a linear relationship with financial inclusion, reflecting different underlying mechanisms.

From Access to Resilience: Financial Inclusion Pathways in Egypt

In the context of this study, SEM was adopted to investigate the pathways through which individual-level socioeconomic characteristics translate into financial resilience. The rationale for building a structural equation model is threefold. First, the study seeks to examine not only the direct determinants of financial inclusion-as captured by account ownership-but also whether financial inclusion itself acts as a channel through which socioeconomic resources shape downstream financial well-being outcomes. Second, SEM allows us to control for multiple covariates across each equation simultaneously, maintaining parsimony while accounting for confounding. Third, the recursive structure of our theoretical model - where income and human capital predict financial access, financial access predicts saving behaviour, and saving behaviour in turn predicts resilience - maps naturally onto the path diagram structure that SEM is designed to estimate.

All indicators in the model are binary (dummy) variables, coded as 0/1. Because all endogenous and exogenous variables are numeric, the Maximum Likelihood (ML) estimator was employed with the Sandwich (robust) standard error correction, known in lavaan as MLR (Yuan-Bentler scaled chi-square).

This combination is consistent with treating binary variables as continuous approximations within a linear probability framework at the structural level, a common and defensible practice when latent variable assumptions are secondary and path interpretability is the primary concern. The Weighted Least Squares Mean and Variance adjusted (WLSMV) estimator - designed specifically for categorical indicators - was initially tested; however, it failed to achieve stable convergence in the present model. Accordingly, Maximum Likelihood (ML) estimation was adopted, and it converged without difficulty.

I. Outcome Variable: Financial Resilience vs. Financial Worry

The choice of the dependent variable is central to any empirical model of financial well-being, especially in survey-based research where consistency across data waves is critical. This study uses data from the 2021 and 2024 waves of the Global Findex Database. However, differences in questionnaire design - particularly regarding financial worry and anxiety - make it difficult to construct a comparable measure of financial stress across both years. Changes in wording, structure, or removal of these items in 2024 introduce measurement inconsistencies, meaning any observed differences over time could reflect either real behavioral changes or survey design effects, thereby weakening the validity of pooled analysis.

To address this issue, the study adopts financial resilience as the main outcome variable. This measure -defined as an individual's ability to access emergency funds within 30 days- remains consistent across both survey waves, ensuring comparability. Beyond its methodological suitability, financial resilience also holds strong theoretical and policy relevance, as it reflects an individual's practical capacity to cope with financial shocks. Accordingly, the binary variable `resilience_num` is used, where 1 indicates the ability to access funds and 0 indicates inability, providing a reliable and meaningful indicator of financial well-being.

II. Exclusion of the Borrowing Variable

Although borrowing was theoretically expected to influence financial resilience, it was tested as a mediating variable and found to be statistically insignificant, leading to its exclusion from the final specification. This likely reflects that the borrowing measure in the Global Findex captures informal practices (such as from family or friends) rather than formal credit, and as shown in Table 2.8, borrowing patterns were similar across account owners and non-owners, limiting its explanatory power..

III. Theoretical Path Structure

The structural equation model estimated in this chapter is grounded in a theoretical pathway from socioeconomic endowment to financial resilience. The conceptual architecture is as follows. Structural socioeconomic factors - income quintile, age, education level, internet access, gender, employment status, survey year, and urbanicity - are posited as exogenous determinants of financial inclusion (account ownership). Financial inclusion, in turn, is hypothesized to facilitate saving behavior by providing a secure vehicle for deposit and accumulation. Finally, saving behavior is theorized as the proximate mechanism through which financial inclusion translates into financial resilience. Additional direct pathways from income, internet access, and education to resilience were also retained, as these relationships were both theoretically plausible and supported by the modification indices suggested by the lavaan output. Accordingly, modification indices were used cautiously as diagnostic tools to refine the model specification while preserving the theoretical coherence of the SEM structure.

This three-stage model is based on well-known ideas in development economics. First, when people have access to bank or mobile accounts, they are more likely to save money. Second, saving-especially for emergencies-is the main way low-income households deal with unexpected financial problems. Third, people who save regularly and have an account are more likely to have emergency funds, meaning they are more financially resilient. So, the model is built on a clear logical pathway, rather than just showing a simple relationship between income and financial resilience. The general SEM form introduced in Methodology is instantiated in the variables of this study through three structural equations, estimated simultaneously. The equations are:

- **Equation 1 (Financial Inclusion):**

$$\begin{aligned} account_num = & \alpha_1 + \beta_1(inc_q_num) + \beta_2(age) + \beta_3(internet_num) + \beta_4(educ_num) \\ & + \beta_5(year) + \beta_6(gender_num) + \beta_7(emp_num) + \beta_8(urban_num) + \varepsilon_1 \end{aligned} \quad (3.5)$$

- **Equation 2 (Saving Behaviour):**

$$\begin{aligned} saved_num = & \alpha_2 + b_1(account_num) + \beta_9(inc_q_num) + \beta_{10}(internet_num) \\ & + \beta_{11}(emp_num) + \beta_{12}(educ_num) + \varepsilon_2 \end{aligned} \quad (3.6)$$

- **Equation 3 (Financial Resilience):**

$$\begin{aligned} resilience_num = & \alpha_3 + d_1(saved_num) + \beta_{13}(account_num) + \beta_{14}(inc_q_num) \\ & + \beta_{15}(internet_num) + \beta_{16}(educ_num) + \varepsilon_3 \end{aligned} \quad (3.7)$$

The parameter b_1 shows how having an account affects saving, while d_1 shows how saving affects financial resilience. When we multiply them ($b_1 \times d_1$), we get the indirect effect of account ownership on financial resilience through saving. This is the main idea of the model. In the R model, this effect is called `ind_account`, and it is tested using a method called the delta method in the lavaan package.

IV. Model Fit

Before interpreting the structural coefficients, it is necessary to evaluate the overall adequacy of the model. Table 3.10 presents the fit statistics for the estimated SEM. The model was estimated in R using the lavaan package (version 0.6-21), with the MLR estimator (Yuan-Bentler scaling correction) to provide robust standard errors appropriate for the binary indicators used.

Table 3.10: Structural Equation Model Fit Statistics

Fit index	Standard	Robust (scaled)
Chi-Square (χ^2)	17.878	18.431
Degrees of Freedom	9	9
P-value	0.037	0.030
CFI	0.991	0.991
TLI	0.973	0.973
RMSEA	0.022	0.023
P(RMSEA $\leq$ 0.05)	0.999	0.999
SRMR	0.010	0.010
Observations (N)	2,002	-

Note. CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardised Root Mean Square Residual. Robust values based on Yuan-Bentler correction.

The model demonstrates excellent fit by conventional psychometric benchmarks. The CFI of 0.991 and TLI of 0.973 both substantially exceed the widely used threshold of 0.95, indicating that the hypothesised path structure accounts for nearly all of the covariance structure in the data. The RMSEA of 0.022 (robust: 0.023) falls well within the range typically considered indicative of close fit (< 0.05), and the p-value for the test of close fit (RMSEA $\leq$ 0.05) is 0.999, confirming that the null hypothesis of acceptable fit cannot be rejected.

The SRMR of 0.010 is exceptionally low, suggesting minimal average residual correlation between observed and model-implied covariances. Together, these indices support the conclusion that the specified three-equation path structure is a well-fitting representation of the data-generating process.

V. Structural Path Coefficients

The estimated structural coefficients for each of the three equations are presented in Tables 2 through 4. Standardised coefficients (Std.all), produced by the lavaan function `standardized = TRUE` using the `std.all` approach, are reported alongside unstandardised estimates. The `std.all` standardisation scales both predictors and outcomes to unit variance, enabling direct comparison of the relative strength of different predictors within and across equations, regardless of their original metric. Effect direction is conveyed by the sign of the coefficient.

Table 3.11: Equation (1) Determinants of Account Ownership (Financial Inclusion)

Predictor	Estimate	Std. Error	Z-value	P-value	Std. All
Income (inc_q_num)	0.114	0.021	5.383	< .001***	0.116
Age	0.011	0.001	17.211	< .001***	0.349
Internet Access (internet_num)	0.122	0.023	5.325	< .001***	0.124
Education (educ_num)	0.187	0.032	5.771	< .001***	0.124
Year (2024)	0.118	0.021	5.632	< .001***	0.121
Gender (gender_num)	0.049	0.024	2.073	.038*	0.050
Employment (emp_num)	0.055	0.024	2.301	.021*	0.056
Urbanicity (urban_num)	-0.013	0.025	-0.515	.606	-0.011
R² (Account Ownership)	-	-	-	-	0.185

The first structural equation examines how socioeconomic and demographic factors affect account ownership. The model explains 18.5% of the variation in account ownership ($R^2 = 0.185$), and most variables are statistically significant except urbanicity. Age is the strongest predictor, showing that older individuals are much more likely to own financial accounts. Education, internet access, and higher income also have strong positive effects, suggesting that financially educated, connected, and wealthier individuals are more likely to participate in the formal financial system.

The survey year variable is also significant, indicating that account ownership increased in 2024 compared to 2021, which reflects the global growth in financial inclusion. Gender and employment status have smaller but still significant positive effects on account ownership. However, urbanicity is not significant, meaning that after controlling for other socioeconomic factors, living in an urban or rural area does not independently affect the likelihood of owning an account in this sample.

Table 3.12: Equation (2) Determinants of Saving Behavior

Predictor	Estimate	Std. Error	Z-value	P-value	Std. All
Account Ownership (b₁)	0.177	0.019	9.255	< .001***	0.208
Income (inc_q_num)	0.082	0.018	4.536	< .001***	0.098
Internet Access (internet_num)	0.111	0.018	6.076	< .001***	0.110
Employment (emp_num)	0.093	0.018	5.106	< .001***	0.110
Education (educ_num)	0.099	0.033	2.957	.003**	0.077
R² (Saving Behaviour)	-	-	-	-	0.130

Note. b₁ denotes the labelled path coefficient for the indirect effect calculation. *** $p < .001$; ** $p < .01$; * $p < .05$.

The second structural equation studies the factors that influence saving behaviour. The model explains 13.0% of the variation in saving behaviour ($R^2 = 0.130$). Account ownership is the strongest predictor of saving and has a highly significant positive effect. This means that people who own formal financial accounts are much more likely to save money, supporting the idea that bank or mobile accounts make saving easier, safer, and more convenient.

Internet access and employment also have important positive effects on saving behaviour. This suggests that digital connectivity helps people use online or mobile saving services, while employment provides income stability that encourages regular saving. Income and education are also significant predictors, although their effects are smaller, indicating that part of their influence on saving may happen indirectly through account ownership.

Table 3.13: Equation (3) Determinants of Financial Resilience

Predictor	Estimate	Std. Error	Z-value	P-value	Std. All
Saving Behaviour (d_1)	0.220	0.026	8.370	< .001***	0.184
Account Ownership	0.037	0.023	1.616	.106	0.036
Income (inc_q_num)	0.196	0.022	6.110	< .001***	0.138
Education (educ_num)	0.099	0.033	3.027	.002**	0.065
R² (Financial Resilience)	-	-	-	-	0.148

Note. d_1 denotes the labelled path coefficient for the indirect effect calculation. *** $p < .001$; ** $p < .01$; * $p < .05$.

The third structural equation examines the factors affecting financial resilience and explains 14.8% of its variation ($R^2 = 0.148$). Saving behaviour is the strongest significant predictor, showing that people who save regularly are more able to handle unexpected financial shocks. Income also has a strong positive effect, meaning wealthier individuals are better able to access emergency funds.

In addition, internet access and education both positively contribute to financial resilience, likely because they improve financial knowledge and access to digital financial services.

On the other hand, the direct effect of account ownership on financial resilience is small and not statistically significant. This means that simply having a financial account does not directly improve a person's ability to deal with financial emergencies. Instead, the main benefit of account ownership comes through encouraging saving behaviour, which then increases financial resilience.

Table 3.14: Indirect Effect of Account Ownership on Financial Resilience via Saving

Effect	Estimate	Std. Error	Z-value	P-value	Std. All
Indirect: Account → Saving → Resilience ($b_1 \times d_1$)	0.039	0.006	6.117	< .001***	0.038

Note. Indirect effect computed as the product of labelled parameters b_1 and d_1 . Standard error derived via the delta method within lavaan.

The formal mediation analysis shows a statistically significant indirect effect of account ownership on financial resilience through saving, with an estimated effect of 0.039 (standardized = 0.038, $z = 6.117$, $p < .001$). In practical terms, holding a formal account is associated with a 3.9 percentage point increase in the predicted probability of being financially resilient via the saving channel. Although the standardized effect size is modest (0.038), it reflects a clear two-step mechanism: account ownership raises the predicted probability of saving by approximately 20.8% of a standard deviation, and saving in turn raises the predicted probability of resilience by about 18.4% of a standard deviation.

These results underscore that financial inclusion through account ownership does not directly guarantee financial resilience; rather, its benefits operate through the behavioral channel of saving. This distinction carries important policy implications: simply expanding account access without complementary interventions to encourage active saving may fail to produce the anticipated improvements in household financial security.

VI. Summary

The chapter uses a three-equation structural equation model estimated with MLR in lavaan to examine how socioeconomic characteristics influence financial inclusion and financial resilience. The model showed excellent fit (CFI = 0.991, TLI = 0.973, RMSEA = 0.022, SRMR = 0.010). Age was the strongest predictor of account ownership, while account ownership was the main predictor of saving behavior. Financial resilience was driven primarily by saving behavior and income, whereas the direct effect of account ownership on resilience was small and non-significant. Mediation analysis confirmed that account ownership improves financial resilience indirectly through increased saving behavior ($b_1 \times d_1 = 0.039$, $p < .001$), indicating that financial inclusion strengthens well-being mainly by promoting precautionary savings. The findings suggest that expanding account access alone is insufficient and should be complemented by policies that encourage regular saving, such as financial literacy programmes, commitment savings products, and digital reminders.

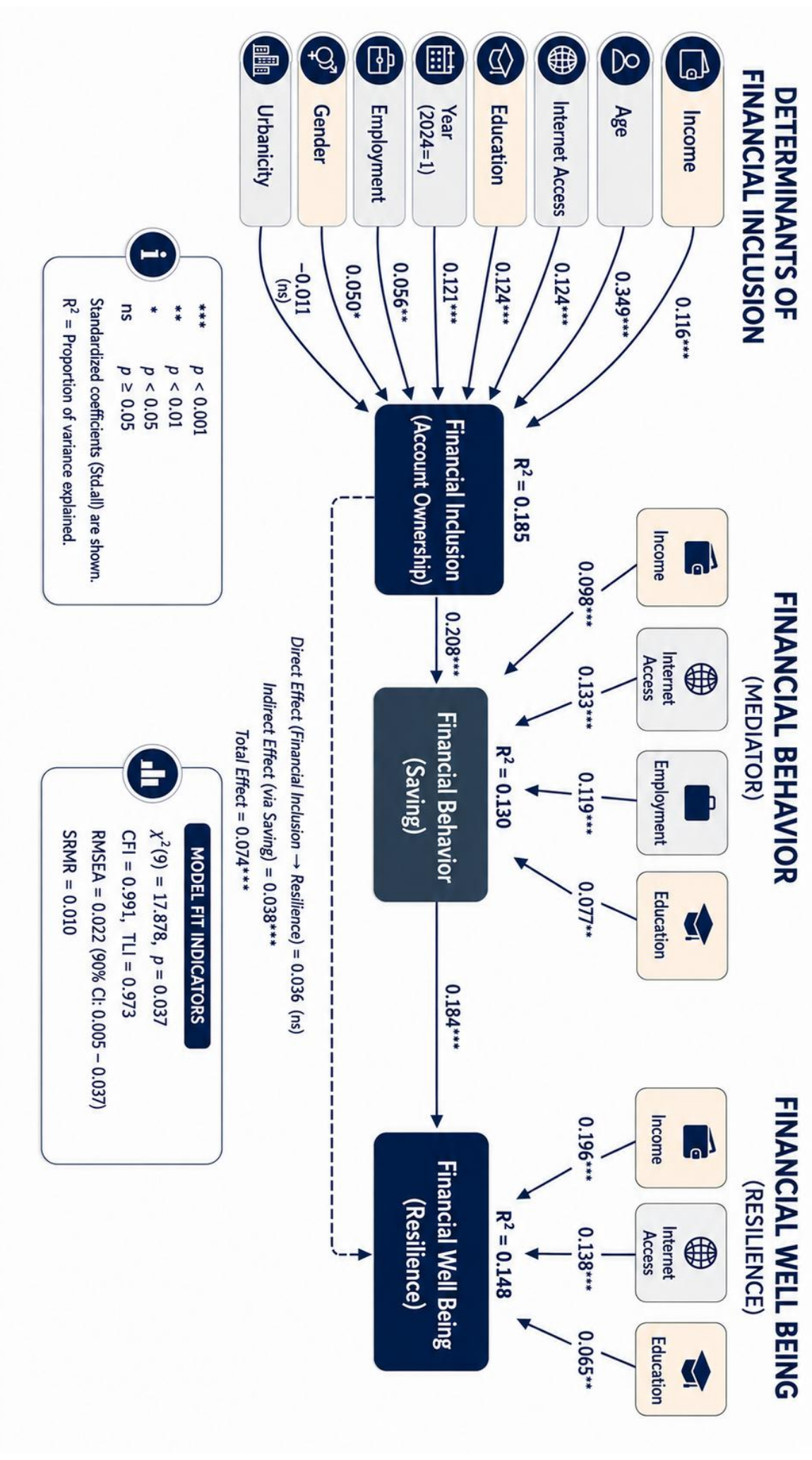


Figure 3.9: Structural Equation Modeling (SEM) framework of financial inclusion and its effects on financial resilience *through savings*

Chapter (4)

Conclusion

Discussion

I. Overview of Financial Inclusion Trends

The findings indicate that financial inclusion in Egypt improved substantially between 2021 and 2024, driven primarily by the expansion of digital financial services and formal financial channels. Account ownership increased from 34.7% to 45.6%, while digital payment usage rose from 25.5% to 38.5%. Mobile money adoption increased markedly from 3.4% to 15.2%, reflecting the growing role of mobile-based financial services, consistent with patterns observed in other developing economies.

The year variable in the access model is positive and statistically significant ($OR = 2.10$), confirming a substantial increase in financial inclusion in 2024 after controlling for socioeconomic factors. This improvement aligns with major digital reforms, including the expansion of Meeza cards, mobile wallets, InstaPay (2022), and the near-complete digitization of government transfers such as “Takaful and Karama” and pensions. Consequently, reliance on cash-based transfers declined sharply from 35.1% to 2.5%.

II. Economic Vulnerability as the Primary Barrier

Economic vulnerability remains the dominant barrier to financial inclusion, reported by nearly 90% of unbanked individuals in both years. Despite digital expansion, insufficient income continues to prevent participation in the formal financial system.

This reflects the broader macroeconomic environment during the study period. Following the COVID-19 pandemic and the Russia - Ukraine conflict, Egypt experienced substantial inflationary pressures and declining household purchasing power, particularly among low- and middle-income groups.

III. The Persistence of Income-Based Inequality (Most Important Finding)

A key finding is the stability of income-based inequality. The interaction between income group and year is not statistically significant ($OR = 1.135$, $p = 0.605$), indicating that the relationship between income and financial inclusion remained unchanged over time. Although overall inclusion increased, the gap between rich and poor persisted.

In substantive terms, while Egyptian government policies successfully increased overall inclusion rates, they failed to alter the underlying relationship between income and financial access. This suggests that achieving equitable financial inclusion requires more than digital infrastructure expansion.

IV. Bidirectional Relationship Between Financial Inclusion and Socioeconomic Position

The results reveal a bidirectional but asymmetric relationship between financial inclusion and socioeconomic status. Higher income significantly increases the likelihood of financial inclusion by 46%, while financial inclusion also increases the probability of higher income, though more weakly ($OR = 1.394$). This suggests that financial inclusion functions more as an outcome of existing socioeconomic advantage than as a strong driver of upward mobility.

V. Education and Internet Access as Determinants of Financial Inclusion

Education and internet access are consistently the most important determinants of financial inclusion across both the logistic regression and SEM results. Education has a strong positive effect, as individuals with tertiary education are about 2.8 times more likely to own a financial account ($OR = 2.775, p < 0.001$), which is also supported by the SEM results ($Std. All = 0.124, p < 0.001$) highlighting the importance of digital literacy.

Internet access is equally important, since individuals without access are around 43% less likely to be financially included ($OR = 0.573, p < 0.001$), with the SEM confirming a similar positive effect ($Std. All = 0.124, p < 0.001$). These results highlight the growing importance of education and digital connectivity as core requirements for financial inclusion in Egypt.

VI. The Widening Urban–Rural Digital Divide

Although internet access improved overall, gains were concentrated in urban areas. Decomposition results show that differences in education, income, and age explain much of the urban - rural gap, indicating that the divide is driven primarily by socioeconomic inequality rather than infrastructure alone. Rural populations continue to face affordability and digital literacy constraints.

A methodological note is important: the observed widening gap may partly reflect improved rural representation in the 2024 sample, making comparisons across years less strictly comparable.

VII. The Digitally Connected Excluded: A New Policy Challenge

A key finding from the clustering analysis is the emergence and rapid growth of a "digitally connected excluded" segment—individuals with internet access but no formal financial account. This group increased from 21.8% of the sample ($N = 219$) in 2021 to 36.6% ($N = 366$) in 2024, becoming the largest cluster. Despite high internet access (86.6% in 2024), only 7.38% engage in digital payments, highlighting extremely low financial engagement. This pattern reveals a clear disconnect between digital connectivity and financial inclusion, indicating that internet access alone is insufficient to ensure meaningful integration into the formal financial system.

VIII. Stagnant Saving and Rising Borrowing

The descriptive analysis reveals an important and potentially worrying trend in financial behavior. While borrowing increased substantially from 43.9% in 2021 to 55.9% in 2024, saving rates remained virtually stagnant at approximately 22% in both waves (22.7% in 2021, 22.3% in 2024).

The bivariate analysis showed a strong and growing association between saving and account ownership (from 53.7% in 2021 to 74.4% in 2024), while borrowing showed only a weak association in both years (37.8% in 2021 and 46.0% in 2024). This suggests that increased borrowing may not necessarily be driven primarily by formal financial inclusion itself but may also reflect broader economic pressures and the continued importance of informal or non-account-based borrowing mechanisms.

IX. The Limited Role of Merchant Digital Payments

Despite overall progress in digital payments, merchant-based digital transactions remained extremely low and virtually unchanged between 2021 (4.4%) and 2024 (4.7%). This suggests that while government transfers and person-to-person payments have digitized successfully, the cash economy remains overwhelmingly dominant for merchant transactions.

X. Evolving Barriers: Documentation and Trust

The share of unbanked individuals citing documentation barriers increased from 6.6% to 10.0%, while trust-related concerns rose from 6.1% to 9.5%. Rather than indicating a deterioration in inclusion conditions, this likely reflects the growing formalization of financial services in Egypt. As digital finance expands, compliance requirements such as KYC procedures and identity verification have become more central to financial access, **making institutional and procedural barriers more visible** as greater segments of the population attempt to enter the formal financial system.

XI. COVID-19 and Behavioral Change: An Exploratory Finding

Among in-store payment users during the pandemic, 71.7% shifted from cash to digital methods, while 28.3% were already digital users. The increased reliance on digital transactions, contactless payments, and mobile-based financial tools appears to have accelerated the normalization of digital financial behavior. For online delivery payments, 62.5% continued to rely on cash on delivery, while 25.0% shifted to online payments and 12.5% were already using digital methods, reflecting partial but incomplete digitalization in this channel.

Nevertheless, the small sample size ($N = 46$ for in-store payments and $N = 24$ for online delivery payments) warrants considerable caution in interpreting these findings.

XII. Gender, Employment, and Financial Inclusion: Women Benefit More from Employment

Employment has a significantly stronger effect on financial inclusion for women than for men, as indicated by the interaction term ($OR = 0.29, p < 0.001$). This implies that formal employment plays a critical role in enabling women's access to financial systems, likely through payroll-linked accounts and increased financial autonomy.

XIII. From Access to Resilience: The Mediating Role of Saving

The structural equation model provides the most theoretically nuanced finding of this study. The model demonstrated excellent fit. The results show that account ownership significantly increases the likelihood of saving, and saving in turn significantly increases financial resilience. However, the direct effect of account ownership on financial resilience was small and not statistically significant. The indirect effect of account ownership on resilience through saving was positive and statistically significant (0.038, 95% bootstrap CI [0.027, 0.053]), but substantively modest - representing only a 3.8 percentage point increase in the probability of resilience.

Which means expanding account access alone is unlikely to meaningfully improve household financial security unless accompanied by interventions that actively encourage formal saving.

Table 4.1: Summary of Empirical Findings Across Methods

Objective	Method Used	Key Insight (Summarized Result)
Determinants and evolution of financial inclusion (2021–2024)	Binary Logistic Regression	Financial inclusion increased significantly following post-2021 reforms, driven mainly by education and income, while gender, employment, and urbanicity were largely insignificant.
Changes in determinants over time	Interaction Logistic Regression	Income-based inequality persisted over time, while gender and employment showed limited or context-dependent effects.
Direction of income–inclusion relationship	Reverse Logistic Regression	Financial inclusion is primarily an outcome of higher income rather than a driver of upward mobility, reinforcing persistent socioeconomic inequality.
Payment behavior post-COVID	Descriptive Analysis	A strong shift toward digital payments is observed for delivery and in-store transactions, while merchant payments remain predominantly cash-based.
Financial inclusion, saving, and resilience	Structural Equation Modeling (SEM)	Financial inclusion improves financial resilience indirectly through saving behavior rather than directly through account ownership.
Financial usage segmentation	K-Modes Clustering	A large “digitally connected but financially excluded” group emerged, highlighting a gap between internet access and actual financial participation.
Urban–rural digital disparities	Fairlie Decomposition	The urban–rural digital gap widened and is primarily explained by income and education rather than geographic location alone.

Final Recommendations

- 1) **Target the "digitally connected excluded":** A large share of individuals have internet access but remain outside the formal financial system. This group should be prioritized through simplified, mobile-friendly financial products that directly convert digital connectivity into financial participation.
- 2) **Simplify account opening for low-income groups:** Economic vulnerability remains the main barrier to financial inclusion. Financial institutions should introduce low-cost or no-fee accounts with minimal balance requirements to make entry into the formal system more accessible for poorer households.
- 3) **Strengthen the role of women’s employment in financial inclusion:** Employment has a stronger impact on financial inclusion for women than for men. Policies should encourage the payment of wages through digital accounts to enhance women’s financial access and autonomy.
- 4) **Promote saving behavior alongside financial access:** Financial inclusion improves resilience mainly through saving rather than account ownership alone. Therefore, financial products should be designed to encourage regular saving through both voluntary and structured mechanisms.
- 5) **Expand and modernize merchant digital payments:** The adoption of digital payments in merchant transactions remains limited. Efforts should focus on increasing merchant acceptance of digital payments through affordable point-of-sale systems and reduced transaction costs for small purchases.

- 6) **Simplify KYC and documentation requirements:** Documentation barriers have become more prominent over time. A tiered KYC system based on simplified identity verification can help reduce exclusion while maintaining regulatory compliance.
- 7) **Build trust in digital financial services:** Trust-related concerns continue to hinder financial participation. Public awareness campaigns and stronger consumer protection frameworks are needed to increase confidence in digital financial systems.
- 8) **Ensure balanced rural representation in data collection:** Accurate measurement of spatial inequality requires consistent and adequate representation of rural populations in national surveys to avoid biased comparisons over time.
- 9) **Monitor risks of rising indebtedness:** While borrowing has increased, saving behavior has remained largely unchanged. This raises concerns about financial vulnerability, highlighting the need for monitoring household debt and promoting responsible borrowing.
- 10) **Strengthen the link between social protection and financial inclusion:** Existing social protection programs have contributed positively to financial inclusion. Expanding their coverage and deepening their integration with digital financial systems can further enhance their impact on inclusion outcomes.

References

- Akerlof, G. A. (1970). The market for “lemons”: Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics*, 84(3), 488–500. <https://www.jstor.org/stable/1879431>
- Agresti, A. (2018). *An introduction to categorical data analysis* (3rd ed.). Wiley.
- Alaa Elden, M. (2024). The role of digital financial inclusion in promoting inclusive growth and environmental sustainability in Egypt. *Scientific Journal for Commercial Research and Studies*, 39(3), 1203–1235. 0.21608/sjrbs.2025.402934.2106
- Allen, F., Demirgüç-Kunt, A., Klapper, L., & Peria, M. S. M. (2016). The foundations of financial inclusion: Understanding ownership and usage of financial accounts. *Journal of Financial Intermediation*, 27, 1–30. <https://www.sciencedirect.com/science/article/abs/pii/S1042957315000534>
- Atkinson, A., & Messy, F. (2013). Promoting financial inclusion through financial education: OECD/INFE evidence, policies and practice. *OECD Working Papers on Finance, Insurance and Private Pensions*, No. 34. OECD Publishing, Paris. <https://doi.org/10.1787/5k3xz6m88smp-en>
- Bollen, K. A. (1989). *Structural equations with latent variables*. Wiley.
- Demirgüç-Kunt, A., Klapper, L., Singer, D., Ansar, S., & Hess, J. (2022). *The Global Findex Database 2021: Financial inclusion, digital payments, and resilience in the age of COVID-19*. World Bank. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099818107072234182>
- Fairlie, R. W. (2005). An extension of the Blinder–Oaxaca decomposition technique to logit and probit models. *Journal of Economic and Social Measurement*, 30(4), 305–316. <https://doi.org/10.3233/JEM-2005-0259>
- Field, A. (2018). *Discovering statistics using IBM SPSS Statistics* (5th ed.). Sage.
- Fu, Jonathan, & Mishra, Mrinal. (2021). Fintech in the time of COVID-19: Technological adoption during crises (November 28, 2021). Swiss Finance Institute Research Paper No. 20-38. <http://dx.doi.org/10.2139/ssrn.3588453>
- Ganna, H., Abdelkhalek, M. A. A., Shokr, A. T. M., & El-Danaf, M. O. M. (2024). Digital financial inclusion and banking stability in Egypt. *Egyptian Journal of Economics and Finance*, 10(17), 295–317. https://csj.journals.ekb.eg/article_342750_0.pdf
- Gower, J. C. (1971). A general coefficient of similarity and some of its properties. *Biometrics*, 27(4), 857–871.
- GSMA. (2026). *The state of the industry report on mobile money 2026 (14th ed.)*. GSMA. <https://www.gsma.com/sotir/>
- Gujarati, D. N., & Porter, D. C. (2009). *Basic econometrics* (5th ed.). McGraw-Hill.
- Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2021). *Multivariate data analysis* (8th ed.). Cengage.
- Harrell, F. E. (2015). *Regression modeling strategies* (2nd ed.). Springer.
- Hegazy, M. S., et al. (2026). Financial inclusion and economic growth in Egypt. *Economics and Organization of Knowledge*, 14(1), 1–18. <https://doi.org/10.2478/eoik-2026-0016>
- Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression* (3rd ed.). Wiley.
- Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis. *Structural Equation Modeling*, 6(1), 1–55.
- Huang, Z. (1998). Extensions to the k-means algorithm for clustering large data sets with categorical values. *Data Mining and Knowledge Discovery*, 2, 283–304.
- International Monetary Fund. (2023). *World Economic Outlook, October 2023: Navigating global divergences*. International Monetary Fund.
- Karlan, D., Ratan, A. L., & Zinman, J. (2014). Savings by and for the poor: A research review and agenda. *Review of Income and Wealth*, 60(1), 36–78. <https://onlinelibrary.wiley.com/doi/10.1111/roiw.12101>

- Kline, R. B. (2016). *Principles and practice of structural equation modeling* (4th ed.). Guilford Press.
- MacKinnon, D. P. (2008). *Introduction to statistical mediation analysis*. Erlbaum.
- Mohamed, A. (2025). Determinants of financial inclusion in Egypt using ML and econometrics. *Journal of Financial and Commercial Research*, 26(3), 1081–1124. https://jsst.journals.ekb.eg/article_433799_0.pdf
- Mohamed, N. G., & Algarhy, F. M. (2024). Assessing the impact of financial technology (Fintech) on financial inclusion in Egypt: An applied study. *International Journal of Accounting and Management Sciences*, 3(3), 433–478. <https://doi.org/10.56830/IJAMS07202410>
- Liu, J., Chen, Y., Chen, X., & Chen, B. (2024). Digital financial inclusion and household financial vulnerability: Evidence from China. *Heliyon*, 10(15), 1–14. <https://doi.org/10.1016/j.heliyon.2024.e35540>
- Sahay, R., et al. (2025). Financial inclusion as a catalyst for economic resilience. *RMD • Economics, Management & Social Sciences*, 2(3), 21–34. <https://doi.org/10.23882/emss25209>
- Shorrocks, A. F. (2013). Decomposition procedures for distributional analysis. *Journal of Economic Inequality*, 11, 99–109.
- Stiglitz, J. E., & Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review*, 71(3), 393–410. <https://www.jstor.org/stable/1802787>
- Wagdi, A. A., & Fathi, R. M. (2025). The impact of digital financial services on banking stability under the moderating role of financial inclusion: Evidence from Egypt. *The Academic Journal of Contemporary Commercial Research*, 5(1), 1–40. 10.21608/ajccr.2025.287891.1109
- Williamson, O. E. (1985). *The economic institutions of capitalism*. Free Press. <https://archive.org/details/economicinstitut0000will>
- Wood, S. N. (2017). *Generalized additive models: An introduction with R* (2nd ed.). Chapman and Hall/CRC.
- Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data* (2nd ed.). MIT Press.
- Yuan, K. H., & Bentler, P. M. (2000). Three likelihood-based methods for mean and covariance structure analysis. *Sociological Methodology*, 30, 165–200.

Appendix A: Data Codebook and Variable Description

This appendix presents the World Bank codebook for the constructed indicators used in the study, including variable definitions for the 2021 and 2024 survey waves. The variables cover demographic characteristics, financial inclusion indicators, digital financial usage, payment behaviors, and access to technology. The coding structure reflects the original survey design and harmonized constructs used in the empirical analysis.

Table 1: Common Constructed Variables (2021 & 2024)

Category	Variable Name	Description
Demographic	female	= 1 if the respondent is female = 2 if the respondent is male
Demographic	age	Respondent's age (in years)
Demographic	educ	= 1 if the respondent has completed primary school or less = 2 if the respondent has completed secondary school = 3 if the respondent has completed tertiary education or more
Demographic	inc_q	Respondent's within-economy household income quintile (1 to 5)
Demographic	emp_in	= 1 if the respondent is in the workforce = 2 if the respondent is out of the workforce
Financial Inclusion	account	= 1 if the respondent has an account at a financial institution, a mobile money account, or both = 0 if the respondent does not have an account
Financial Inclusion	account_fin	= 1 if the respondent had an account at a bank or at another type of financial institution, such as a credit union, a microfinance institution, a cooperative, or the post office (if applicable), or has a debit card = 0 if the respondent did not have an account
Financial Inclusion	account_mob	= 1 if the respondent used mobile money services to pay bills or to send or receive money in the past year = 0 if the respondent did not use them
Financial Inclusion	saved	= 1 if the respondent personally saved or set aside money in the past year, including using an account at a financial institution, via a mobile money account, a savings club or person outside the family, for any reason = 0 if the respondent did not save
Financial Inclusion	borrowed	= 1 if the respondent, personally or together with someone else, borrowed money in the past year, including from a bank or similar financial institution, via a mobile money account, from family or friends, or from an informal savings group, or for any other reason = 0 if the respondent did not borrow

Digital Financial Usage	anydigpayment	= 1 if respondent used mobile money, a card, or a mobile phone to make a payment from an account, or used the internet to pay bills or to buy something online or in a store, or paid bills or sent or received remittances directly from or into a financial institution account or through a mobile money account in the past year. It also includes respondents who received payments for agricultural products, government transfers, wages, or a public sector pension into a financial institution account or through a mobile money account in the past year. = 0 if the respondent did not make or receive a digital payment
Digital Financial Usage	merchantpay_dig	=1 if respondent used a card, or a mobile phone, to make a purchase in-store or to pay online for an internet purchase in the past year =0 if respondent did not make a digital merchant payment
Payment & Income Flows	receive_wages	= 1 if the respondent received a wage payment into an account in the past year = 2 if the respondent received a wage payment in cash only in the past year = 3 if the respondent received a wage payment via some other method in the past year = 4 if the respondent did not receive a wage payment in the past year = 5 if the respondent does not know/refuses to answer
Payment & Income Flows	receive_transfers	= 1 if the respondent received government transfers into an account in the past year = 2 if the respondent received government transfers in cash only in the past year = 3 if the respondent received government transfers via some other method in the past year = 4 if the respondent did not receive government transfers in the past year = 5 if the respondent does not know/refuses to answer
Payment & Income Flows	receive_agriculture	= 1 if the respondent received payment from the sale of agricultural products into an account in the past year = 2 if the respondent received payment from the sale of agricultural products in cash only in the past year = 3 if the respondent received payment from the sale of agricultural products via some other method in the past year
Payment & Income Flows	pay_utilities	= 1 if the respondent made a utility payment from an account in the past year = 2 if the respondent made a utility payment in cash only in the past year = 3 if the respondent made a utility payment via some other method in the past year = 4 if the respondent did not make a utility payment in the past year = 5 if the respondent does not know/refuses to answer

Table 2: 2021-Specific Variables

Category	Variable Name	Description
Demographic	urbanicity_f2f	= 1 if the respondent lives in a rural area = 2 if and the respondent lives in an urban area

Payment & Income Flows	receive_pension	= 1 if the respondent received pension from the government into an account = 2 if the respondent received pension from the government in cash only = 3 if the respondent received pension from the government via some other method = 4 if the respondent did not receive pension from the government = 5 if the respondent does not know/refuses to answer
Payment & Income Flows	remittances	= 1 if the respondent sent or received domestic remittances via an account = 2 if the respondent sent or received domestic remittances through an MTO service = 3 if the respondent sent or received domestic remittances in cash only = 4 if the respondent sent or received domestic remittances via other methods = 5 if the respondent did not send or receive domestic remittances = 6 if the respondent does not know/refuses to answer
Technology Access	internetaccess	= 1 if the respondent has access to the Internet =2 if no =3 if don't know =4 if refused to answer
Technology Access	mobileowner	= 1 if the respondent owns a mobile phone =2 if no =3 if don't know =4 if refused to answer

Table 3: 2024-Specific Variables

Category	Variable Name	Description
Demographic	urbanicity	= 1 if the respondent lives in a rural area = 2 if and the respondent lives in an urban area
Financial Inclusion	dig_account	=1 if the respondent has a mobile money account or an account at a bank or similar financial institution that they make digital payments from using a card or phone =0 if the respondent does not have a mobile money account or an account at a bank or similar financial institution that they make digital payments from using a card or phone
Payment & Income Flows	receive_pensions	= 1 if the respondent received pension from the government into an account in the past year = 2 if the respondent received pension from the government in cash only in the past year = 3 if the respondent received pension from the government via some other method in the past year = 4 if the respondent did not receive pension from the government in the past year = 5 if the respondent does not know/refuses to answer

Payment & Income Flows	domestic_remittances	= 1 if the respondent sent or received domestic remittances via an account in the past year = 2 if the respondent sent or received domestic remittances via other methods in the past year = 3 if the respondent did not send or receive domestic remittances in the past year = 4 if the respondent does not know/refuses to answer
Technology Access	internet_use	=1 if respondent has used the internet in the past three months =0 if respondent has not used the internet in the past three months
con1	Have a mobile phone	=1 if respondent has a mobile phone that they use to make and receive personal calls =2 if no =3 if don't know =4 if refused

Appendix B: Model Specification and Diagnostic Tests

Table 1: Testing proportional odds assumption for income quantile (reversed model)

Variable	Chi-square (X ²)	df	p-value	Assumption Result
Omnibus Test	65.58	24	0.000	Violated
accountYes	6.91	3	0.070	Borderline
age	11.16	3	0.010	Violated
internetaccessNo	0.74	3	0.860	Holds
genderMale	3.79	3	0.290	Holds
educSecondary	4.11	3	0.250	Holds
educTertiary_or_more	10.35	3	0.020	Violated
emp_inIn_workforce	4.82	3	0.190	Holds
urbanicityUrban	3.42	3	0.330	Holds

The proportional odds assumption was assessed using the Brant test following the ordered logistic regression estimation. The omnibus test indicated a violation of the parallel regression assumption, suggesting that some explanatory variables exert non-uniform effects across the ordinal categories of the dependent variable. Specifically, age and tertiary education exhibited statistically significant violations of the proportional odds assumption.

Additionally, the original income quintile variable displayed instability and non-parallel effects across response categories. To address this issue and improve model robustness, income groups were collapsed into broader categories. This regrouping reduced category sparsity and enhanced the stability and interpretability of the estimated coefficients while preserving the ordinal structure of the income variable.

Appendix C: Structural Equation Modelling Estimation and Model Assumptions

This appendix presents the estimation framework, estimator selection, and identification assumptions used in the Structural Equation Modelling (SEM) analysis.

C.1 Estimation Method

The model is estimated using Structural Equation Modelling (SEM) within a full-information maximum likelihood framework. Parameters are obtained by maximizing the fit between the sample covariance matrix S and the model-implied covariance matrix $\Sigma(\theta)$:

$$F_{ML} = \log |\Sigma(\theta)| + \text{tr}(S\Sigma^{-1}(\theta)) - \log |S| - p$$

where p denotes the number of observed variables.

Model refinement is guided by modification indices based on the Lagrange Multiplier principle, which are used as diagnostic tools for identifying potential improvements in model specification while maintaining theoretical consistency.

C.2 Treatment of Categorical Variables and Estimator Choice

Although the model includes categorical (binary) observed variables, the Weighted Least Squares Mean and Variance adjusted (WLSMV) estimator was initially considered. However, WLSMV did not achieve stable convergence under the specified recursive structure.

Such convergence issues are common in complex SEM models involving multiple endogenous variables and mediation pathways. Therefore, the analysis proceeds using robust Maximum Likelihood estimation (MLR), which provides a quasi-maximum likelihood solution and adjusts standard errors using a sandwich (Huber–White) estimator together with a Yuan–Bentler scaled chi-square correction. This estimator is robust to non-normality and is widely used in applied SEM when the focus is on structural relationships rather than latent measurement modelling (Bollen, 1989; Kline, 2016; Yuan & Bentler, 2000).

C.3 Model Identification and Assumptions

The model is recursive and therefore identified under standard SEM conditions, with unidirectional causal ordering and no feedback loops. The key assumptions include:

- Correct specification of structural relationships
- Linearity of relationships
- Independence of observations
- Adequate sample size for stable estimation

Note: This appendix complements the main text by providing technical details on estimation procedures and underlying modelling assumptions used in the SEM framework.

Appendix D: Spline Specification in Logistic Regression Models

This appendix presents the mathematical formulation of the spline-based logistic regression specification used in the main analysis to capture potential nonlinear relationships between continuous explanatory variables and a binary outcome.

D.1 General Logistic Regression with Spline Terms

To allow for flexible nonlinear effects, spline functions are incorporated within the logistic regression framework. The general specification is:

$$\text{logit}(P(Y_i = 1)) = \beta_0 + f(x_i) + Z_i'\gamma$$

where $P(Y_i = 1)$ denotes the probability of the binary outcome, Z_i is a vector of control variables, and $f(x_i)$ represents a smooth nonlinear transformation of the continuous predictor x_i , approximated using spline basis functions (Harrell, 2015; Wood, 2017).

D.2 Cubic Spline Specification

A commonly used functional form is the cubic spline representation:

$$f(x_i) = \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \sum_{k=1}^K \theta_k (x_i - \kappa_k)_+^3$$

where:

- κ_k are knot locations
- $(x_i - \kappa_k)_+ = \max(0, x_i - \kappa_k)$
- K denotes the number of knots

D.3 Restricted Cubic Spline (Natural Spline)

In applied econometric analysis, the restricted cubic spline (natural spline) is often preferred due to its flexibility in the central range and imposed linearity in the tails of the distribution. This improves model stability and interpretability while maintaining nonlinear flexibility (Harrell, 2015; Wood, 2017).

Note: The spline specification is used to reduce functional form misspecification and to capture nonlinear relationships between continuous predictors and the binary outcome within the logistic regression framework.

Appendix E: Mathematical Formulation of the Fairlie Decomposition Framework

This appendix presents the mathematical formulation of the nonlinear decomposition framework developed by Fairlie (2005), which is used in the main analysis to decompose differences in binary outcome probabilities between groups.

E.1 Nonlinear Decomposition Framework

The Fairlie decomposition extends the Oaxaca–Blinder decomposition to nonlinear models such as logit and probit specifications. The mean difference in predicted probabilities between two groups is expressed as:

$$\bar{Y}_A - \bar{Y}_B = \left[\frac{1}{N_A} \sum_{i=1}^{N_A} F(X_i^A \hat{\beta}^A) - \frac{1}{N_B} \sum_{i=1}^{N_B} F(X_i^B \hat{\beta}^A) \right] + \left[\frac{1}{N_B} \sum_{i=1}^{N_B} F(X_i^B \hat{\beta}^A) - \frac{1}{N_B} \sum_{i=1}^{N_B} F(X_i^B \hat{\beta}^B) \right]$$

where $F(\cdot)$ represents the cumulative distribution function of the logistic or probit model.

E.2 Variable-Level Decomposition

The marginal contribution of each explanatory variable is obtained using a sequential replacement approach. For variable k under permutation r , the contribution is defined as $\Delta_{k,r}$. The average contribution of variable k across all permutations is given by:

$$\hat{C}_k = \frac{1}{R} \sum_{r=1}^R \Delta_{k,r}$$

E.3 Symmetric Decomposition Adjustment

To reduce sensitivity to the choice of reference group, a symmetric decomposition is applied by estimating the decomposition in both directions ($A \rightarrow B$ and $B \rightarrow A$). The final symmetric contribution is computed as:

$$C_k^{sym} = \frac{C_k^{A \rightarrow B} + C_k^{B \rightarrow A}}{2}$$

E.4 Resampling Procedure

To improve robustness, a resampling-with-replacement procedure is implemented. The decomposition is repeatedly estimated over multiple bootstrap samples, and the final estimates are obtained as:

$$\bar{C}_k = \frac{1}{B} \sum_{b=1}^B C_k^{(b)}$$

where B denotes the number of bootstrap replications.

Note: All equations presented in this appendix correspond to the nonlinear decomposition framework introduced by Fairlie (2005) and are implemented in the main empirical analysis.